

Derived categories and Brauer groups of log del Pezzo surfaces with $\frac{1}{3}(1, 1)$ singularities

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Abstract

We study the derived categories of log del Pezzo surfaces with only $\frac{1}{3}(1, 1)$ singularities. For the Fano index one families classified by Corti and Heuberger, we construct explicit full exceptional collections on their canonical stacks using blow-up presentations of the minimal resolutions and cascades of smooth points. More generally, for any log del Pezzo surface X with k such singularities, we obtain the length formula $12 - K_X^2 + 4k/3$ independent of the Fano index. We show that the intersection lattice of the exceptional curves determines the Brauer group: it vanishes for $k \leq 5$ and is isomorphic to $\mathbb{Z}/3$ for $k = 6$. Applying the Karmazyn–Kuznetsov–Shinder framework, we obtain semiorthogonal decompositions of the bounded derived categories of coherent sheaves on the singular surfaces for the one- and three-point cascades, together with a nontrivial Brauer-twisted decomposition for the six-point cascade. For quasismooth degree-10 hypersurfaces in $\mathbb{P}(1, 2, 3, 5)$, we construct a thirteen-object full exceptional collection on the canonical stack, compute its Euler matrix, and obtain a semiorthogonal decomposition of the derived category of the singular surface. We also identify a general such hypersurface as the anticanonical model of the blow-up of $\mathbb{P}(1, 1, 3)$ at eight general smooth points, with the converse holding for general centers. Finally, for rational surfaces with arbitrary baskets of $\frac{1}{n_i}(1, 1)$ singularities, we give a general block formula for the Euler matrices of full exceptional collections on their canonical stacks.

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1 Introduction

Del Pezzo surfaces offer a particularly explicit setting in which mirror symmetry interacts with the birational classification of Fano varieties. The Fukaya–Seidel formalism for Lefschetz fibrations [31] gives the symplectic framework for homological mirror symmetry with Landau–Ginzburg mirrors. An early stacky instance of this picture was given by Auroux, Katzarkov, and Orlov [5], who proved homological mirror symmetry for weighted projective planes, treating their generally singular coarse spaces through smooth quotient stacks. Ueda [33] proved the correspondence for smooth toric del Pezzo surfaces,

and Auroux, Katzarkov, and Orlov [4] established it for del Pezzo surfaces obtained by blowing up $\mathbb{P}^2$. Ueda and Yamazaki [34] subsequently extended this picture to certain quotient stacks of toric del Pezzo surfaces.

A complementary mirror-symmetric perspective relates deformation families, Fano polygons, and periods for the same surfaces. In the smooth case, the ten deformation families correspond to the ten mutation-equivalence classes of Fano polygons with empty residual basket, and the classical periods of the corresponding maximally mutable Laurent polynomials recover their regularized quantum periods [2, 24]. More generally, Akhtar et al. [2] formulated a mirror-symmetric classification program relating mutation classes of Fano polygons to $\mathbb{Q}$ -Gorenstein deformation classes of locally $\mathbb{Q}$ -Gorenstein rigid del Pezzo surfaces with cyclic quotient singularities that admit $\mathbb{Q}$ -Gorenstein toric degenerations.

The simplest non-smooth residual singularity in this program is $\frac{1}{3}(1, 1)$. This $\mathbb{Q}$ -Gorenstein rigid singularity coincides with its residue in the sense of singularity content [2, 1]. Surfaces with only these singularities provide a first testing ground for extending the smooth theory: they exhibit new orbifold phenomena while remaining accessible to explicit classification. Corti and Heuberger classified them into 29 $\mathbb{Q}$ -Gorenstein deformation families and constructed explicit models; precisely 26 of these families admit $\mathbb{Q}$ -Gorenstein degenerations to toric surfaces [11].

For these families, Kasprzyk, Nill, and Prince [24] classified the corresponding Fano polygons up to mutation. Oneto and Petracci [28] computed many of the quantum periods and compared them with classical periods of associated Laurent polynomials, using conjectural extensions of Quantum Lefschetz and the Abelian/non-Abelian Correspondence to the orbifold setting.

In keeping with this stacky B -model perspective, the orbifold invariants in this picture are naturally defined on the canonical smooth Deligne–Mumford stack $\pi: \mathcal{X} \rightarrow X$ of a log del Pezzo surface X [2, 28]. This stack records the stabilizer data at the quotient singularities, while its bounded derived category of coherent sheaves gives the categorical B -model. We ask how the contribution of a $\frac{1}{3}(1, 1)$ point to this category relates to the exceptional curve on the minimal resolution, and what categorical decompositions descend to X .

To address these questions, let $\nu: \tilde{X} \rightarrow X$ be the minimal resolution. We compare

$$D^b(\mathrm{coh} \tilde{X}), \quad D^b(\mathrm{coh} \mathcal{X}), \quad \text{and} \quad D^b(\mathrm{coh} X)$$

by constructing exceptional collections on $\tilde{X}$ and using them to obtain semiorthogonal decompositions on the canonical stack $\mathcal{X}$ and, under suitable descent conditions, on the singular surface X itself. We carry out these constructions along the Corti–Heuberger cascades.

Ishii–Ueda’s special McKay correspondence [21] gives a fully faithful functor $\Phi: D^b(\mathrm{coh} \tilde{X}) \rightarrow D^b(\mathrm{coh} \mathcal{X})$. For singularities of type $\frac{1}{n_i}(1, 1)$, its semiorthogonal complement is generated by $n_i - 2$ exceptional objects supported on the residual gerbe at each singular point. For any rational surface with quotient singularities, Orlov’s blow-up formula yields a full exceptional collection on its minimal resolution. Ishii–Ueda’s result then induces such a collection on the canonical stack, as observed by Gugiatti–Rota [17, Remark 4.8] and recalled in Proposition 3.5. Canonical stacks of log del Pezzo surfaces admit full exceptional collections, since their resolutions are rational (Proposition 2.2).

To pass to the singular surface, Karmazyn–Kuznetsov–Shinder (KKS) introduced a framework to construct semiorthogonal decompositions of $D^b(\mathrm{coh} X)$ from suitable

decompositions on a resolution [23, Theorems 2.12 and 3.16]. A decomposition on $\tilde{X}$ that is compatible with the contraction descends to X . Under the stronger adherence condition, which prescribes the form of the components associated with contracted curves, the components corresponding to contracted exceptional curves become equivalent to bounded derived categories of modules over finite-dimensional local algebras.

For surfaces with only $\frac{1}{3}(1, 1)$ singularities, general results guarantee full exceptional collections on their canonical stacks and determine their Brauer group orders via the Corti–Heuberger defect [11, Section 2.1, Proposition 18(b), Lemma 28]. Blow-up presentations of the minimal resolutions of the six rigid bases, together with Orlov’s formula, give explicit exceptional collections. We identify the Brauer groups through saturation of the exceptional lattice, including the alternating class for Z_6 , and construct untwisted KKS decompositions for the one- and three-point cascades and a Brauer-twisted decomposition for the six-point cascade. For X_{10} , we relate general hypersurfaces to eight general points on $\mathbb{P}(1, 1, 3)$. The Euler-matrix formula extends to arbitrary baskets of $\frac{1}{n_i}(1, 1)$ points.

The constructions below provide explicit B -model data—exceptional collections, Euler pairings, and coarse-space components—for the $\frac{1}{3}(1, 1)$ families appearing in the mirror-symmetric classification program.

Every Fano index one family $X_{k,d}$, indexed by the number k of its singular points and by the degree $d = K_X^2$, belongs to one of six cascades whose members arise from a rigid base Z_k by blowing up $K_{Z_k}^2 - d$ smooth points [11, Theorems 4 and 6, Corollary 8]. Here $Z_1 = \mathbb{P}(1, 1, 3)$, while for $2 \leq k \leq 6$, Z_k is the unique surface in its family; Table 1 lists the invariants.

In a marked blow-up presentation, the marking records the ordered centers, their incidence data, and the curves contracted to singular points (Definition 5.3). The starting smooth surfaces Y_0 and collection lengths N_k are listed in Table 1. Write $\mathcal{A}_k = \langle A_1^{(k)}, \dots, A_{N_k}^{(k)} \rangle$ for the exceptional collection on $\tilde{Z}_k$ (Proposition 5.4, Appendix C). The presentations of $\tilde{Z}_k$ may involve infinitely near points, whereas the additional blow-up centers on Z_k are distinct by ampleness, as explained after Definition 5.3.

The following theorem gives full exceptional collections on the canonical stacks of surfaces in the Fano index one families. Here the residual-gerbe objects are the exceptional objects supported at the stacky singular points.

Theorem 1.1 (Explicit collections in the Corti–Heuberger cascades). *Let $X \in X_{k,d}$ be a member of a Fano index one Corti–Heuberger family, with singular points $p_1, \dots, p_k$, and let $X \rightarrow Z_k$ be a blow-up at $s = K_{Z_k}^2 - d$ distinct smooth points of the rigid base. Let $\gamma: \tilde{X} \rightarrow \tilde{Z}_k$ be the induced map of minimal resolutions, with exceptional curves $D_1, \dots, D_s$. For the base collection $\mathcal{A}_k = \langle A_1^{(k)}, \dots, A_{N_k}^{(k)} \rangle$, the ordered collection*

$$\langle \mathcal{O}_{D_s}(-1), \dots, \mathcal{O}_{D_1}(-1), L\gamma^* A_1^{(k)}, \dots, L\gamma^* A_{N_k}^{(k)} \rangle$$

is a full exceptional collection in $D^b(\mathrm{coh} \tilde{X})$. Its image under the Ishii–Ueda embedding Φ , preceded by the residual-gerbe objects $\mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_k}$, is a full exceptional collection in $D^b(\mathrm{coh} X)$ of length

$$N_k + s + k = 12 - d + \frac{4k}{3}.$$

Appendix C constructs marked blow-up presentations of the six base resolutions, and Corollary C.3 extends their Orlov collections along the Corti–Heuberger cascades.

For these rational surfaces, Bright’s exact sequence [9] yields

$$\mathrm{Br}(X) \cong \mathrm{coker}(\theta: \mathrm{Pic} \tilde{X} \rightarrow \Lambda^\vee),$$

where $\Lambda \subset \text{Pic } \tilde{X}$ is the sublattice generated by the curves contracted by ν . The map θ sends a divisor class to its intersection pairing with these curves (Corollary 4.9). The saturation calculation in Appendix B then identifies this cokernel with the Pontryagin dual of the saturation quotient:

$$\text{Br}(X) \cong \text{Hom}(\Lambda^{\text{sat}}/\Lambda, \mathbb{Q}/\mathbb{Z}), \quad \text{where} \quad \Lambda^{\text{sat}} = (\Lambda \otimes \mathbb{Q}) \cap \text{Pic } \tilde{X}.$$

Because $\Lambda^{\text{sat}}/\Lambda \cong \text{Cl}(X)_{\text{tors}}$, the Brauer group is also the dual of the torsion subgroup of the divisor class group (Lemma B.2). Since an untwisted adherent KKS decomposition forces $\text{Cl}(X)$ to be torsion-free [23, Corollary 4.8], a nonzero Brauer group immediately obstructs such a decomposition. Invariance of the Brauer group under blow-ups at smooth points (Lemma B.5) reduces its computation to the six rigid bases and gives the following theorem.

Theorem 1.2 (Brauer obstruction in the Corti–Heuberger cascades). *Let X belong to a Corti–Heuberger cascade with k singular points, all of type $\frac{1}{3}(1, 1)$. Then*

$$\text{Br}(X) \cong \begin{cases} 0, & 1 \leq k \leq 5, \\ \mathbb{Z}/3, & k = 6. \end{cases}$$

Among the six cascades, only the six-point case has a Brauer obstruction to an untwisted adherent decomposition in the sense of KKS.

Appendix B computes these intersection maps via toric divisor relations and configurations of curves forming k -gons, explicitly exhibiting the alternating generator

$$\frac{1}{3}(E_1 - E_2 + E_3 - E_4 + E_5 - E_6) \in \Lambda^{\text{sat}}/\Lambda$$

for Z_6 . The bases Z_k are toric precisely when $k \in \{1, 3, 6\}$ [11, Tables 1, 2 and 4], [28, Table 1]. On their minimal resolutions, we construct subcategories adherent to the curves contracted to singular points—incorporating a nontrivial Brauer twist in the six-point case. Because compatibility and adherence are preserved under additional blow-ups away from these curves (Lemma 4.4), we obtain semiorthogonal decompositions of $D^b(\text{coh } X)$ for the one- and three-point cascades (Corollaries 5.11 and B.6), and of $D^b(\text{coh}(X, \alpha))$ for the six-point cascade, with a nonzero class $\alpha \in \text{Br}(X)$ (Corollary B.8).

For $k \in \{2, 4, 5\}$, the Brauer group vanishes. Although necessary, this vanishing does not by itself produce compatible decompositions with simultaneous adherent subcategories. These non-toric bases lack the requisite torus action, and their toric degenerations acquire additional singularities (Remark B.7).

Quasismooth degree-10 hypersurfaces $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$ belong to the family $X_{1,1/3}$. Their exceptional collections and derived category decompositions follow from our cascade results. We single out this family to compare its hypersurface parameter space with that of configurations of eight smooth points on $\mathbb{P}(1, 1, 3)$, proving that general hypersurfaces correspond to eight general smooth centers and vice versa (part (iii) of the theorem below).

Theorem 1.3 (The hypersurface X_{10}). *Let $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$ be a quasismooth hypersurface of degree 10, with canonical stack $\mathcal{X}_{10}$ and minimal resolution $\nu: \tilde{X}_{10} \rightarrow X_{10}$. Then X_{10} has a unique singular point p , of type $\frac{1}{3}(1, 1)$, and $\tilde{X}_{10}$ is the blow-up $\sigma: \tilde{X}_{10} \rightarrow \mathbb{F}_3$ at eight distinct points off C_0 , with pairwise disjoint exceptional curves $D_1, \dots, D_8$.*

(i) $D^b(\mathrm{coh} \mathcal{X}_{10})$ admits the full exceptional collection

$$\langle \mathcal{E}_p, \Phi(\mathcal{O}_{D_1}(-1)), \dots, \Phi(\mathcal{O}_{D_8}(-1)), \Phi(\sigma^*\mathcal{O}), \Phi(\sigma^*\mathcal{O}(f)), \\ \Phi(\sigma^*\mathcal{O}(C_0 + 3f)), \Phi(\sigma^*\mathcal{O}(C_0 + 4f)) \rangle$$

of length 13, whose Euler matrix is computed in [Proposition 5.9](#).

(ii) The singular surface admits a semiorthogonal decomposition

$$D^b(\mathrm{coh} X_{10}) = \langle D^b(K(3, 1)\text{-mod}), F_1, \dots, F_{10} \rangle, \\ K(3, 1) \cong \mathbb{C}[z_1, z_2]/(z_1, z_2)^2,$$

with exceptional objects F_i made explicit in [Theorem 5.10](#).

(iii) A general such hypersurface is the anticanonical model of the blow-up of $\mathbb{P}(1, 1, 3)$ at eight general smooth points, and conversely general centers give a general hypersurface ([Propositions 5.7](#) and [A.6](#)).

[Lemma 5.5](#) establishes the unique singularity, while Corti–Heuberger [[11](#), Corollary 8(1)] gives the specified blow-up presentation at eight smooth points; their distinctness follows from ampleness, as explained after [Definition 5.3](#). Part (i) specializes [Theorem 1.1](#) to $k = 1$ and $s = 8$. Part (ii) follows from the KKS decomposition of $\mathbb{P}(1, 1, 3)$ [[23](#), Example 3.17], with compatibility and adherence preserved under the eight blow-ups ([Corollary 4.6](#)). Part (iii) combines the Reid–Suzuki construction [[30](#)] with the dominance of the classifying map for the hypersurfaces obtained from eight-point configurations, proved in [Appendix A](#).

The X_{10} case fits into the broader investigation of log del Pezzo hypersurfaces and their derived categories. Johnson–Kollár [[22](#)] classified anticanonically embedded log del Pezzo hypersurfaces in weighted projective spaces. On the categorical side, Elagin [[15](#)] and Gugiatti–Rota [[17](#)] constructed exceptional collections for series of log del Pezzo hypersurfaces different from the degree-ten family considered here.

Finally, we give a uniform block upper-triangular formula for the Euler matrices of full exceptional collections on canonical stacks of rational surfaces with arbitrary baskets of $\frac{1}{n_i}(1, 1)$ singularities. The diagonal blocks record the local singularity contributions and the Euler matrix on the resolution; when the resolution collection consists of line bundles, the off-diagonal blocks are determined by their intersections with the exceptional curves ([Proposition 6.1](#)). For X_{10} , the matrix is a specialization of this formula. We also obtain uniform collection-length formulas ([Corollary 3.6](#)), with the $\frac{1}{3}(1, 1)$ case given in [Corollary 3.7](#).

[Sections 2](#) to [4](#) recall exceptional collections on resolutions, the Ishii–Ueda embedding, and KKS descent. [Section 5](#) constructs the collections and decompositions for the cascades and X_{10} , and [Section 6](#) computes the Euler matrices. [Appendix A](#) compares the parameter spaces of eight-point configurations and degree-10 hypersurfaces, proving that general centers give general hypersurfaces and conversely. [Appendix B](#) computes the Brauer groups, and [Appendix C](#) gives explicit blow-up presentations of the minimal resolutions of the six rigid bases.

Notation and conventions

All varieties and stacks are defined over $\mathbb{C}$. We write $\mathbb{F}_n$ for the n -th Hirzebruch surface, C_0 for its negative section, and f for its fiber class. The notation $\mathbb{P}(E)$ denotes the

Grothendieck projectivization parametrizing one-dimensional quotients of E . We write $\nu : \tilde{X} \rightarrow X$ for the minimal resolution of the singular surface under consideration; other letters are reserved for auxiliary blow-ups and contractions. In the general theory, $r = \# \text{Sing}(X)$ and p_i is of type $\frac{1}{n_i}(1, 1)$; for one point we write n . In the Corti–Heuberger notation $X_{k,d}, Z_k$, we have $n_i = 3$ and $r = k$. We distinguish resolution curves E_i over the singular points, blow-up curves D_i over smooth centers, and local objects $\mathcal{E}_{p_i,j}$, defined in Definition 3.3; for a $\frac{1}{3}(1, 1)$ -point we write $\mathcal{E}_{p_i} = \mathcal{E}_{p_i,2}$.

2 Resolutions and exceptional collections

To construct full exceptional collections on canonical stacks in Section 3, we first produce them on the minimal resolutions of log del Pezzo surfaces. As recalled in Proposition 2.2 below, the minimal resolution $\tilde{X}$ of a log del Pezzo surface X is rational. Successively contracting (-1) -curves yields a minimal rational surface, hence $\mathbb{P}^2$ or $\mathbb{F}_n$ with $n \neq 1$ [6, Theorem V.10]. Reversing these contractions exhibits $\tilde{X}$ as an iterated blow-up of that model. Orlov’s blow-up formula transports the standard collection on the starting surface to the resolution and proves Corollary 2.7.

We begin with the surface geometry, recalling the rationality argument before turning to exceptional collections. The relevant background on log del Pezzo surfaces can be found in [3], and the rationality statement is also recalled in [17, Remark 4.3]. Our treatment of exceptional collections on smooth rational surfaces follows [19].

Definition 2.1. A *log del Pezzo surface* is a normal projective surface X with Kawamata log terminal (klt) singularities such that $-K_X$ is ample.

For normal surfaces, klt singularities are precisely quotient singularities [25, Proposition 4.18]. They are also rational [25, Theorem 5.22]. If $\nu : \tilde{X} \rightarrow X$ is the minimal resolution, then $\nu_*\mathcal{O}_{\tilde{X}} = \mathcal{O}_X$ and $R^i\nu_*\mathcal{O}_{\tilde{X}} = 0$ for all $i > 0$. The Leray spectral sequence gives a canonical isomorphism

$$H^1(\tilde{X}, \mathcal{O}_{\tilde{X}}) \cong H^1(X, \mathcal{O}_X).$$

We recall the rationality argument.

Proposition 2.2. *Let X be a log del Pezzo surface, and let $\nu : \tilde{X} \rightarrow X$ be its minimal resolution. The surface $\tilde{X}$ is rational. In particular, X is rational.*

Proof. Since X is klt and $-K_X$ is ample, Kawamata–Viehweg vanishing for klt varieties [25, Theorem 2.70] gives $H^1(X, \mathcal{O}_X) = 0$. The rationality of the singularities and the Leray isomorphism above then give $H^1(\tilde{X}, \mathcal{O}_{\tilde{X}}) = 0$.

It remains to show that $H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(2K_{\tilde{X}})) = 0$. Suppose, to the contrary, that there exists an effective divisor $D \sim 2K_{\tilde{X}}$. The pullback $-\nu^*K_X$ is nef because $-K_X$ is ample, so $(-\nu^*K_X) \cdot D \geq 0$. On the other hand, write the discrepancy formula as

$$K_{\tilde{X}} = \nu^*K_X + \sum_i a_i E_i.$$

The projection formula gives $\nu^*K_X \cdot E_i = 0$ for every ν -exceptional curve E_i and $(\nu^*K_X)^2 = K_X^2$. Intersecting the discrepancy formula with $-\nu^*K_X$ therefore gives

$$(-\nu^*K_X) \cdot D = (-\nu^*K_X) \cdot 2K_{\tilde{X}} = -2K_X^2 < 0.$$

This contradiction implies $H^0(\tilde{X}, \mathcal{O}_{\tilde{X}}(2K_{\tilde{X}})) = 0$. By Castelnuovo’s rationality criterion, $\tilde{X}$ is rational. Since X is birational to $\tilde{X}$, the surface X is rational as well. $\square$

Example 2.3. The weighted projective plane $\mathbb{P}(1, 1, 3)$ has a unique singularity of type $\frac{1}{3}(1, 1)$. Its minimal resolution is $\mathbb{F}_3 = \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(3))$, whose negative section C_0 , with $C_0^2 = -3$, is the exceptional locus [23, Example 3.17].

We next recall the exceptional collections and blow-up formula used below.

Definition 2.4. Let $\mathcal{T}$ be a triangulated category. An object $G \in \mathcal{T}$ is *exceptional* if $\text{Hom}(G, G) = \mathbb{C}$ and $\text{Ext}^k(G, G) = 0$ for all $k \neq 0$. An ordered sequence $\langle G_1, \dots, G_n \rangle$ is an *exceptional collection* if each G_i is exceptional and $\text{Ext}^\bullet(G_j, G_i) = 0$ for all $j > i$. The collection is *full* if the smallest full triangulated subcategory of $\mathcal{T}$ containing $G_1, \dots, G_n$ is $\mathcal{T}$.

The projective line has the standard full exceptional collection

$$D^b(\text{coh } \mathbb{P}^1) = \langle \mathcal{O}, \mathcal{O}(1) \rangle,$$

and the projective plane has Beilinson's full exceptional collection [7]

$$D^b(\text{coh } \mathbb{P}^2) = \langle \mathcal{O}, \mathcal{O}(1), \mathcal{O}(2) \rangle.$$

Orlov's projective bundle formula [29, Corollary 2.7], applied to

$$\mathbb{F}_n = \mathbb{P}_{\mathbb{P}^1}(\mathcal{O} \oplus \mathcal{O}(n))$$

gives a full exceptional collection on $\mathbb{F}_n$. The intersection numbers are $C_0^2 = -n$, $C_0 \cdot f = 1$, and $f^2 = 0$, and the relative tautological line bundle is

$$\mathcal{O}_{\mathbb{F}_n}(1) \simeq \mathcal{O}(C_0 + nf).$$

Orlov's projective-bundle formula gives

$$D^b(\text{coh } \mathbb{F}_n) = \langle p^* D^b(\text{coh } \mathbb{P}^1), p^* D^b(\text{coh } \mathbb{P}^1) \otimes \mathcal{O}_{\mathbb{F}_n}(1) \rangle,$$

where $p : \mathbb{F}_n \rightarrow \mathbb{P}^1$ is the ruling. For every $s \in \mathbb{Z}$, $\langle \mathcal{O}_{\mathbb{P}^1}(s-n), \mathcal{O}_{\mathbb{P}^1}(s-n+1) \rangle$ is a full exceptional collection of $D^b(\text{coh } \mathbb{P}^1)$. Using the standard pair in the first component and this shifted pair in the second, together with $\mathcal{O}(C_0 + sf) = \mathcal{O}_{\mathbb{F}_n}(1) \otimes p^* \mathcal{O}_{\mathbb{P}^1}(s-n)$, gives

$$\langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + sf), \mathcal{O}(C_0 + (s+1)f) \rangle, \quad s \in \mathbb{Z}, \quad (1)$$

a full exceptional collection for every value of s ; the case $s = n$ recovers the standard collection.

Theorem 2.5 (Orlov's blow-up formula [29, Theorem 4.3 and Corollary 4.4]). *Let S be a smooth projective surface, and let $\beta : \text{Bl}_p S \rightarrow S$ be the blow-up of S at a point p , with exceptional curve D . There is a semiorthogonal decomposition*

$$D^b(\text{coh } \text{Bl}_p S) = \langle \mathcal{O}_D(-1), L\beta^* D^b(\text{coh } S) \rangle,$$

where $\mathcal{O}_D(-1)$ is regarded as an object of $D^b(\text{coh } \text{Bl}_p S)$ via pushforward along the inclusion $D \hookrightarrow \text{Bl}_p S$. In particular, if $D^b(\text{coh } S)$ admits a full exceptional collection, the category $D^b(\text{coh } \text{Bl}_p S)$ also admits one.

Corollary 2.6 ([19, Theorem 5.6]). *Every smooth projective rational surface admits a full exceptional collection.*

Proof. Express the surface as an iterated point blow-up of $\mathbb{P}^2$ or a Hirzebruch surface $\mathbb{F}_n$ [6, Theorem V.10]. By successive applications of Theorem 2.5, the standard full exceptional collection on the starting surface extends to a full exceptional collection on the given surface, with one exceptional object added at each step. $\square$

Together, Proposition 2.2 and Corollary 2.6 yield the following consequence.

Corollary 2.7. *Let X be a log del Pezzo surface, and let $\nu : \tilde{X} \rightarrow X$ be its minimal resolution. The category $D^b(\text{coh } \tilde{X})$ admits a full exceptional collection.*

3 Canonical stacks and exceptional collections

This section introduces canonical stacks and the local components in the special McKay correspondence for singularities of type $\frac{1}{n_i}(1, 1)$, with $n_i \geq 2$. The global decomposition yields the existence statement of Proposition 3.5, while the discrepancy and rank computations give Corollary 3.6. We then specialize to $\frac{1}{3}(1, 1)$ -singularities and compute the local adjoint needed for the Euler matrices.

Definition 3.1. Let X be a normal surface with quotient singularities. The *canonical stack* of X is the smooth Deligne–Mumford stack $\pi : \mathcal{X} \rightarrow X$ whose coarse moduli space is X , whose local models at quotient singularities are the corresponding quotient stacks, and whose structure morphism is an isomorphism over the smooth locus of X . Existence for quotient singularities follows from Vistoli’s construction [35, Proposition 2.8 and its proof]. Uniqueness up to isomorphism over X follows from the universal property in [16, Theorem 4.6 and Remark 4.9].

At a singular point of type $\frac{1}{n}(1, 1)$, with $n \geq 2$, the local analytic model is $\mathbb{A}^2/\mu_n$, and the canonical stack is locally $[\mathbb{A}^2/\mu_n]$. We use the ring-action convention $\rho_i(\zeta) = \zeta^i$, with the coordinate functions $x, y \in \mathbb{C}[x, y]$ both of weight ρ_1 . All character labels below use this ring-action convention.

For the cyclic quotient $\mathbb{A}^2/\mu_n$, let $R = \mathbb{C}[x, y]$ and let $\tau : Y \rightarrow \text{Spec } R^{\mu_n}$ be the minimal resolution. For a character ρ , set $M_\rho := (R \otimes \rho^\vee)^{\mu_n}$, and let its associated *full sheaf* on Y be $\mathcal{M}_\rho := \tau^* M_\rho / \text{torsion}$. These constructions associate indecomposable full sheaves on Y with characters of μ_n [21, Section 2]. Wunram calls such a sheaf *special* when $H^1(Y, \mathcal{M}_\rho^\vee) = 0$. Equivalently, ρ is a *special representation*; otherwise it is called *non-special*. This terminology follows Wunram [36] and the exposition in [21, Section 2].

Lemma 3.2. *Let X be a normal surface with singular points $p_1, \dots, p_r$, where p_i has type $\frac{1}{n_i}(1, 1)$, with $n_i \geq 2$. At p_i , the special representations of μ_{n_i} are precisely ρ_0 and ρ_1 . The non-special representations are $\rho_2, \dots, \rho_{n_i-1}$, so there are $n_i - 2$ of them. For $n_i = 2$ this list is empty; for $n_i = 3$ it consists of ρ_2 alone.*

Proof. At each point, the Hirzebruch–Jung continued fraction $n_i/1 = [n_i]$ gives the recursion $j_0 = n_i$, $j_1 = 1$, and $j_2 = j_0 - n_i j_1 = 0$. Wunram’s cyclic special McKay correspondence identifies the entries preceding the terminal zero with the special character labels, read modulo n_i [21, Theorems 2.3 and 3.2]. These give $\rho_{j_0} = \rho_{n_i} = \rho_0$ and $\rho_{j_1} = \rho_1$; all remaining characters are non-special [17, Example 2.6]. $\square$

Definition 3.3. Let X be a normal surface with singular points $p_1, \dots, p_r$ of types $\frac{1}{n_i}(1, 1)$, with $n_i \geq 2$, and let $\mathcal{X}$ be its canonical stack. For the residual gerbe $\iota_{p_i} : B\mu_{n_i} \hookrightarrow \mathcal{X}$, define the *local exceptional objects*

$$\mathcal{E}_{p_i, j} := \iota_{p_i*}(\mathcal{O}_{B\mu_{n_i}} \otimes \rho_j), \quad 2 \leq j \leq n_i - 1,$$

and their ordered exceptional collection

$$\mathcal{B}_{p_i} := \langle \mathcal{E}_{p_i,2}, \dots, \mathcal{E}_{p_i,n_i-1} \rangle.$$

The collection is empty when $n_i = 2$. Locally, $\mathcal{E}_{p_i,j}$ is represented by the equivariant module $\mathbb{C}[x,y]/(x,y) \otimes \rho_j$, namely $\mathcal{O}_0 \otimes \rho_j$ in the notation of Ishii–Ueda. For a point p of type $\frac{1}{3}(1,1)$, we retain the abbreviation $\mathcal{E}_p := \mathcal{E}_{p,2}$.

Theorem 3.4 (Ishii–Ueda [21, Theorems 1.4 and 3.1]). *Let X be a surface with at worst quotient singularities, let $\mathcal{X}$ be its canonical stack, and let $Y \rightarrow X$ be the minimal resolution. There is a fully faithful functor*

$$\Phi : D^b(\mathrm{coh} Y) \rightarrow D^b(\mathrm{coh} \mathcal{X})$$

and a semiorthogonal decomposition

$$D^b(\mathrm{coh} \mathcal{X}) = \langle \mathcal{E}_1, \dots, \mathcal{E}_q, \Phi(D^b(\mathrm{coh} Y)) \rangle,$$

where $\mathcal{E}_1, \dots, \mathcal{E}_q$ form an exceptional collection.

If the singularities are $p_1, \dots, p_r$ of types $\frac{1}{n_i}(1,1)$, with $n_i \geq 2$, the local description [21, Theorem 3.1] and [17, Example 2.6] gives

$$D^b(\mathrm{coh} \mathcal{X}) = \langle \mathcal{B}_{p_1}, \dots, \mathcal{B}_{p_r}, \Phi(D^b(\mathrm{coh} Y)) \rangle,$$

with the collections of Definition 3.3 and $q = \sum_{i=1}^r (n_i - 2)$. Collections at distinct points are mutually orthogonal and can be ordered arbitrarily; within each collection the character indices increase from 2 to $n_i - 1$.

We fix the ordering of [21, Theorem 1.4], with the exceptional objects preceding the image of Φ . Thus $\mathrm{RHom}(\Phi A, \mathcal{E}_i) = 0$, which gives the lower-left zero blocks in the Euler matrices.

The following existence statement was observed in [17, Remark 4.8].

Proposition 3.5 (Existence on rational quotient surfaces). *Let X be a normal projective rational surface with quotient singularities, and let $\mathcal{X}$ be its canonical stack. Then $D^b(\mathrm{coh} \mathcal{X})$ admits a full exceptional collection. In particular, this holds for every log del Pezzo surface.*

Proof. The minimal resolution of X is smooth and rational, so Corollary 2.6 supplies a full exceptional collection. Insert it into the Ishii–Ueda decomposition of Theorem 3.4. The log del Pezzo case follows from Proposition 2.2. $\square$

For the one-curve cyclic singularities, the local collections also determine an ordered collection and its length.

Corollary 3.6 (Mixed-basket length and ordered collection). *Let X be a normal projective rational surface with singular points $p_1, \dots, p_r$ of types $\frac{1}{n_i}(1,1)$, with $n_i \geq 2$. Let $\mathcal{X}$ be its canonical stack and $\nu : \tilde{X} \rightarrow X$ its minimal resolution. With the local collections $\mathcal{B}_{p_i}$ of Definition 3.3, for any full exceptional collection $\langle G_1, \dots, G_N \rangle$ on $\tilde{X}$, the Ishii–Ueda embedding gives the ordered full exceptional collection*

$$\langle \mathcal{B}_{p_1}, \dots, \mathcal{B}_{p_r}, \Phi(G_1), \dots, \Phi(G_N) \rangle$$

on $\mathcal{X}$. The lengths of the resolution and stack collections are, respectively,

$$N = \operatorname{rk} K_0(\tilde{X}) = 12 - K_X^2 + \sum_{i=1}^r \frac{(n_i - 2)^2}{n_i},$$

$$\ell = N + \sum_{i=1}^r (n_i - 2) = 12 - K_X^2 + \sum_{i=1}^r \frac{2(n_i - 1)(n_i - 2)}{n_i}.$$

Proof. The resolution has a full exceptional collection by [Corollary 2.6](#). Write $E_i \subset \tilde{X}$ for the exceptional curve over p_i . Thus $E_i^2 = -n_i$, and the discrepancy formula is

$$K_{\tilde{X}} = \nu^* K_X - \sum_{i=1}^r \frac{n_i - 2}{n_i} E_i.$$

To verify the coefficient of E_i , write it as a_i and intersect with E_i . Adjunction gives $K_{\tilde{X}} \cdot E_i = -2 - E_i^2 = n_i - 2$, whereas $\nu^* K_X \cdot E_i = 0$. Hence $-n_i a_i = n_i - 2$, giving $a_i = -(n_i - 2)/n_i$. The curves E_i are disjoint, so

$$K_{\tilde{X}}^2 = K_X^2 - \sum_{i=1}^r \frac{(n_i - 2)^2}{n_i}.$$

Noether's formula [[18](#), Appendix A, Example 4.1.2] gives $12\chi(\mathcal{O}_{\tilde{X}}) = K_{\tilde{X}}^2 + c_2(\tilde{X})$. For the smooth rational surface $\tilde{X}$ ([Proposition 2.2](#)), we have $\chi(\mathcal{O}_{\tilde{X}}) = 1$ and $c_2(\tilde{X}) = 2 + \rho(\tilde{X})$, the latter being its topological Euler characteristic, so $\rho(\tilde{X}) = 10 - K_{\tilde{X}}^2$. Hence

$$N = \operatorname{rk} K_0(\tilde{X}) = 2 + \rho(\tilde{X}) = 12 - K_{\tilde{X}}^2 = 12 - K_X^2 + \sum_{i=1}^r \frac{(n_i - 2)^2}{n_i}.$$

Inserting the resolution collection into [Theorem 3.4](#) gives the displayed collection, with $n_i - 2$ additional objects at each point. Its length is

$$\ell = N + \sum_{i=1}^r (n_i - 2) = 12 - K_X^2 + \sum_{i=1}^r \frac{2(n_i - 1)(n_i - 2)}{n_i}. \quad \square$$

For the singularities used in the Corti–Heuberger cascades, the local collections consist of single exceptional objects. Setting $n_i = 3$ in [Corollary 3.6](#) gives the following specialization.

Corollary 3.7. *Let X be a log del Pezzo surface whose singularities are $p_1, \dots, p_r$, all of type $\frac{1}{3}(1, 1)$. Let $\mathcal{X}$ be its canonical stack, and let $\tilde{X} \rightarrow X$ be its minimal resolution. Then $D^b(\operatorname{coh} \mathcal{X})$ admits a full exceptional collection of length $\ell = 12 - K_X^2 + \frac{4r}{3}$. The resolution collection has length $N = \operatorname{rk} K_0(\tilde{X}) = 12 - K_X^2 + \frac{r}{3}$, and each local collection in [Corollary 3.6](#) is the singleton $\mathcal{E}_{p_i} = \mathcal{E}_{p_i, 2}$.*

The preceding existence and length results use only the Ishii–Ueda semiorthogonal decomposition. For the Euler matrices we also need the opposite pairings $\operatorname{RHom}(\mathcal{E}_i, \Phi A)$. In the projective setting, Φ has both adjoints: the semiorthogonal ordering above is controlled by the right adjoint, which kills the local objects, while the left adjoint Ψ computes them through $\operatorname{RHom}(\mathcal{E}_i, \Phi A) \simeq \operatorname{RHom}(\Psi(\mathcal{E}_i), A)$. We now fix the local conventions needed to compute this left adjoint.

With the convention $\rho_i(\zeta) = \zeta^i$ and x, y of weight ρ_1 , the equivariant Koszul resolution of $\mathcal{O}_0 \otimes \rho_i$ has free terms $\mathcal{O} \otimes \rho_i$, $(\mathcal{O} \otimes \rho_{i+1})^{\oplus 2}$, and $\mathcal{O} \otimes \rho_{i+2}$ in degrees $0, -1, -2$, respectively.

On $\mathcal{U}_3 = [\mathbb{A}^2/\mu_3]$, the form $dx \wedge dy$ has character ρ_2 , so $\omega_{\mathcal{U}_3} \simeq \mathcal{O}_{\mathbb{A}^2} \otimes \rho_2$, whereas the anticanonical bundle has character ρ_1 [17, Section 3, equation (17)]. The local character of $\mathcal{O}_{\mathcal{X}_{10}}(1)$, the anticanonical bundle in our hypersurface example, is ρ_1 , not ρ_2 . This distinction fixes the character convention in the calculation below and in the Euler pairings of Section 6.

Lemma 3.8 (Local left adjoint for $\frac{1}{n}(1, 1)$). *Let $n \geq 3$, let $Y = \text{Tot}_{\mathbb{P}^1} \mathcal{O}_{\mathbb{P}^1}(-n)$, with projection $p : Y \rightarrow \mathbb{P}^1$ and zero section E , and let ψ be the left adjoint of the local Ishii–Ueda embedding ϕ . Then*

$$\psi(\mathcal{O}_0 \otimes \rho_i) \simeq \begin{cases} 0, & 0 \leq i \leq n-3, \\ \mathcal{O}_E(-n)[1], & i = n-2, \\ \mathcal{O}_E(-n+1), & i = n-1. \end{cases} \quad (2)$$

Here $\mathcal{O}_E(t)$ has degree t on $E \simeq \mathbb{P}^1$.

Proof. With our character convention, the formula follows from [17, Example 3.5, equation (26)] after translating the indices. We spell out this translation, together with the Koszul calculation, to make the character residues and cohomological shifts explicit. In the notation of [17, Theorem 3.1], one has $\rho_{\text{nat}} = \rho_1^{\oplus 2}$, $\rho_{\text{det}} = \rho_2$, and ρ_1 is the nontrivial special character. Thus $\rho_{\text{det}}^\vee = \rho_{n-2}$, and the condition $\rho_j \otimes \rho_{\text{det}} = \rho_1$ gives $j = n-1$. The dual appearing in $M_\rho = (\mathbb{C}[x, y] \otimes \rho^\vee)^{\mu_n}$ acts on the representation used to form the module; it does not change the label of the equivariant skyscraper $\mathcal{O}_0 \otimes \rho$. Passing to the opposite geometric indexing convention negates every character index modulo n . Although only the characters $2 \leq j \leq n-1$ occur in the exceptional complement, we carry out the computation for all characters to display the convention completely.

Put $P_a = p^* \mathcal{O}_{\mathbb{P}^1}(a)$ for every $a \in \mathbb{Z}$, and $\mathcal{L}_a = \mathcal{O}_{\mathbb{A}^2} \otimes \rho_a$. The $\mathcal{L}_a$ are periodic modulo n , whereas the line bundles P_a are not. In particular, P_{a+n} cannot be identified with P_a , so character residues and line-bundle degrees must be kept separate. The weight-one quotient of $\mathbb{C}x \oplus \mathbb{C}y$ over $\mathbb{P}^1$ is $\mathcal{O}_{\mathbb{P}^1}(1)$, and its powers yield the tautological character summands P_a for $0 \leq a < n$. The description of the tautological bundles associated with the universal μ_n -cluster over Y in [17, proof of Theorem 3.1, equations (20)–(22)] therefore gives $\psi(\mathcal{L}_a) = P_{-a}$ in exactly this range.

We now apply ψ to the equivariant Koszul resolution

$$0 \longrightarrow \mathcal{L}_{j+2} \xrightarrow{(-y, x)} \mathcal{L}_{j+1}^{\oplus 2} \xrightarrow{(x, y)} \mathcal{L}_j \longrightarrow \mathcal{O}_0 \otimes \rho_j \longrightarrow 0.$$

Let s_0, s_1 be the homogeneous coordinates on $\mathbb{P}^1$, pulled back to Y , and let $t \in H^0(Y, P_{-n})$ be the section cutting out E . Write $u = (-s_1, s_0)^\top$ and $v = (s_0, s_1)$. The character indices must first be reduced modulo n , and only then may $\mathcal{L}_a$ be replaced by P_{-a} . With this order of operations, the resulting complexes in degrees $-2, -1, 0$ are

$$\begin{aligned} 0 \leq j \leq n-3 : & \quad [P_{-j-2} \xrightarrow{u} P_{-j-1}^{\oplus 2} \xrightarrow{v} P_{-j}], \\ j = n-2 : & \quad [P_0 \xrightarrow{tu} P_{1-n}^{\oplus 2} \xrightarrow{v} P_{2-n}], \\ j = n-1 : & \quad [P_{-1} \xrightarrow{u} P_0^{\oplus 2} \xrightarrow{tv} P_{1-n}]. \end{aligned}$$

Here the factor t records multiplication across the character boundary from $n-1$ to 0 ; crucially, it has degree $-n$, not degree zero. The first complex is the exact Euler

sequence and hence has no cohomology. For the second complex, $\ker v = P_{-n}$, and its only cohomology is the quotient $P_{-n}/tP_0 = \mathcal{O}_E(-n)$ in degree -1 . For the third, $\operatorname{coker} u = P_1$, while the induced map $P_1 \xrightarrow{t} P_{1-n}$ is injective with cokernel $\mathcal{O}_E(1-n)$ in degree zero. These three cases give (2), including the stated shifts. $\square$

For several singular points the calculation applies componentwise. The restriction $n \geq 3$ here concerns only the non-special objects in the semiorthogonal complement of $\phi(D^b(\operatorname{coh} Y))$; the KKS descent theorem used in Section 4 also allows $n = 2$.

4 The singular surface and the KKS approach

We now study $D^b(\operatorname{coh} X)$ for the singular surface X . For a single $\frac{1}{n}(1,1)$ -point, a decomposition on $\mathbb{F}_n$ compatible with contraction of its negative section descends by the Karmazyn–Kuznetsov–Shinder (KKS) theorem [23, Theorems 2.12 and 3.16]; adherence identifies the component corresponding to the contracted curve. Lemma 4.4 proves that compatibility and adherence are preserved under successive point blow-ups whose centers avoid the strict transforms of the contracted curves, including infinitely near centers. The Corti–Heuberger application in Section 5 uses $n = 3$.

KKS descent requires a decomposition on the resolution compatible with the contraction. The theorem assumes rational singularities, a hypothesis satisfied here by the klt property, as recalled in Section 2.

We begin with the compatibility condition that permits descent from a resolution.

Definition 4.1 (KKS compatibility, [23, Definition 2.7]). Let $\tau : \tilde{X} \rightarrow X$ be a resolution of a normal projective surface with rational singularities, viewed as a birational contraction, and let

$$D^b(\operatorname{coh} \tilde{X}) = \langle \tilde{\mathcal{A}}_1, \dots, \tilde{\mathcal{A}}_t \rangle$$

be a semiorthogonal decomposition. The decomposition is *compatible with the contraction* τ if, for every irreducible curve E contracted by τ , the object $\mathcal{O}_E(-1)$ belongs to one of the components $\tilde{\mathcal{A}}_i$.

Theorem 4.2 (KKS descent, [23, Theorem 2.12(i)]). *In the setting of Definition 4.1, a compatible decomposition descends uniquely to*

$$D^b(\operatorname{coh} X) = \langle \mathcal{A}_1, \dots, \mathcal{A}_t \rangle, \quad \mathcal{A}_i = R\tau_*(\tilde{\mathcal{A}}_i).$$

Let $\mathcal{D}_i$ be the union of the exceptional curves assigned to $\tilde{\mathcal{A}}_i$. The restriction of $R\tau_$ induces an equivalence with the Verdier quotient*

$$\mathcal{A}_i \simeq \tilde{\mathcal{A}}_i / \langle \mathcal{O}_E(-1) \mid E \subset \mathcal{D}_i \rangle.$$

Here E runs over the irreducible components of $\mathcal{D}_i$. In particular, pushforward is an equivalence on a component with no assigned curves. The descended projection functors have finite cohomological amplitude. Both assertions are contained in KKS Theorem 2.12(i).

To identify the descended components, we need the stronger condition of adherence. In the one-curve case used here, its untwisted form has the following concrete description.

Definition 4.3 (One-curve case of [23, Definition 3.6]). Let S be a smooth projective surface such that $H^\bullet(S, \mathcal{O}_S) \cong \mathbb{C}$, and let $E \subset S$ be a smooth rational curve with $E^2 = -n$, where $n \geq 2$. A triangulated subcategory $\mathcal{A} \subset D^b(\text{coh } S)$ is *untwisted adherent* to E if there is a line bundle L_0 such that

$$\mathcal{A} = \langle L_0, \mathcal{O}_E(-1) \rangle, \quad L_0 \cdot E = n - 1.$$

By [23, Lemma 3.5, Definition 3.6, and Example 3.7], this is equivalent to a presentation

$$\mathcal{A} = \langle L_0, L_1 \rangle, \quad L_1 = L_0(E), \quad L_0 \cdot E = n - 1.$$

More generally, b -twisted adherence replaces $\mathcal{O}_E(-1)$ by $\mathcal{O}_E(-1 + b)$ and the intersection condition by $L_0 \cdot E = n + b - 1$. The one-point decompositions below use $b = 0$; nonzero twists are needed for the Brauer-twisted six-point construction in Appendix B.4.

The following lemma proves preservation of compatibility and adherence under blow-up for all the contracted curves simultaneously.

Lemma 4.4. *Let S be a smooth projective surface and let $C_1, \dots, C_s \subset S$ be pairwise disjoint smooth rational curves with $C_i^2 = -n_i$, where $n_i \geq 2$. Let $\beta : S' \rightarrow S$ be the blow-up of a point outside $\bigcup_i C_i$, and write C'_i for the strict transforms. If a semiorthogonal decomposition of $D^b(\text{coh } S)$ contains components $\mathcal{A}_i$ such that*

$$\mathcal{O}_{C_i}(-1) \in \mathcal{A}_i,$$

then $\mathcal{O}_{C'_i}(-1) \in L\beta^ \mathcal{A}_i$ for every i . More generally, if $H^\bullet(S, \mathcal{O}_S) \cong \mathbb{C}$ and subcategories $\mathcal{A}_i$ have b_i -twisted adherent presentations*

$$\mathcal{A}_i = \langle L_i, L_i(C_i) \rangle, \quad L_i \cdot C_i = n_i + b_i - 1,$$

then every $L\beta^ \mathcal{A}_i$ is b_i -twisted adherent to C'_i and contains $\mathcal{O}_{C'_i}(b_i - 1)$. In particular, untwisted adherence is preserved when all $b_i = 0$. These conclusions hold after any finite sequence of point blow-ups whose centers avoid the successive strict transforms of all the C_i , even when a center is a point infinitely near to an earlier blow-up center.*

Proof. The calculation for a single curve can be carried out independently for each of the pairwise disjoint curves C_i . Since β is an isomorphism near every C_i , for any integer b_i we have

$$L\beta^* \mathcal{O}_{C_i}(b_i - 1) \simeq \mathcal{O}_{C'_i}(b_i - 1), \quad \beta^* C_i = C'_i.$$

For the line bundles,

$$\beta^*(L_i(C_i)) = (\beta^* L_i)(C'_i), \quad (\beta^* L_i) \cdot C'_i = L_i \cdot C_i = n_i + b_i - 1.$$

A point blow-up preserves the cohomology of the structure sheaf, and derived pullback is fully faithful by Orlov's formula. Each pullback component therefore remains adherent, with the same integer b_i . Induction gives the conclusions for any finite sequence. The same argument applies to infinitely near centers, since each blow-up remains an isomorphism near the successive strict transforms of the C_i . The first pullback identity with $b_i = 0$ proves compatibility independently of adherence. $\square$

For one cyclic point, the following proposition states the KKS decomposition of $\mathbb{P}(1, 1, n)$ from [23, Example 3.17] in our notation.

Proposition 4.5 (KKS weighted-projective-plane decomposition). *Let $n \geq 2$, and let $\tau : \mathbb{F}_n \rightarrow \mathbb{P}(1, 1, n)$ contract the negative section C_0 , with $C_0^2 = -n$. Writing f for the fiber class, the full exceptional collection*

$$\langle \mathcal{O}(-C_0 - f), \mathcal{O}(-f), \mathcal{O}, \mathcal{O}(C_0 + nf) \rangle$$

begins with generators of an untwisted adherent subcategory $\tilde{\mathcal{A}}_0 = \langle \mathcal{O}(-C_0 - f), \mathcal{O}(-f) \rangle$ over C_0 . It descends to

$$D^b(\text{coh } \mathbb{P}(1, 1, n)) = \langle D^b(K(n, 1)\text{-mod}), \mathcal{O}, \mathcal{O}(n) \rangle,$$

where the Kalck–Karmazyn algebra is

$$K(n, 1) \cong \mathbb{C}[z_1, \dots, z_{n-1}] / (z_1, \dots, z_{n-1})^2.$$

Proof. The collection is the projective-bundle collection used in [23, Example 3.17], with their exceptional section and fiber class denoted here by C_0 and f . Fullness follows directly from (1) with $s = n + 1$, which gives

$$\langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + (n + 1)f), \mathcal{O}(C_0 + (n + 2)f) \rangle.$$

Tensoring by $\mathcal{O}(-f)$ yields

$$\langle \mathcal{O}(-f), \mathcal{O}, \mathcal{O}(C_0 + nf), \mathcal{O}(C_0 + (n + 1)f) \rangle.$$

Since

$$\omega_{\mathbb{F}_n} = \mathcal{O}(-2C_0 - (n + 2)f),$$

the Serre functor satisfies

$$S(\mathcal{O}(C_0 + (n + 1)f)) = \mathcal{O}(-C_0 - f)[2].$$

The Serre transform of the last object in a full exceptional collection may be moved cyclically to the front, giving

$$\langle \mathcal{O}(-C_0 - f)[2], \mathcal{O}(-f), \mathcal{O}, \mathcal{O}(C_0 + nf) \rangle.$$

Shifting an individual exceptional object preserves semiorthogonality and does not change the triangulated subcategory it generates, so the first object may be replaced by $\mathcal{O}(-C_0 - f)$. The divisor sequence

$$0 \longrightarrow \mathcal{O}(-C_0 - f) \longrightarrow \mathcal{O}(-f) \longrightarrow \mathcal{O}_{C_0}(-1) \longrightarrow 0$$

puts $\mathcal{O}_{C_0}(-1)$ in $\tilde{\mathcal{A}}_0$, proving compatibility. The equality $(-C_0 - f) \cdot C_0 = n - 1$ proves untwisted adherence. The surface is rational with rational singularities, so KKS [23, Theorem 2.12(i)] descends this compatible decomposition. By untwisted adherence, [23, Theorem 3.16] identifies the first descended component with $D^b(K(n, 1)\text{-mod})$. The remaining components descend equivalently to the line bundles $\mathcal{O}$ and $\mathcal{O}(n)$, since $\tau^*\mathcal{O}(n) = \mathcal{O}(C_0 + nf)$. [23, Example 3.14(2)] gives the displayed presentation of $K(n, 1)$. $\square$

The KKS decomposition of $\mathbb{P}(1, 1, n)$ above comes from an adherent subcategory on $\mathbb{F}_n$. Its derived pullback remains adherent under successive blow-ups away from the strict transforms of C_0 , by Lemma 4.4, giving the following decomposition.

Corollary 4.6 (One-point descent under blow-ups). *Let X be a normal projective surface with a unique singularity of type $\frac{1}{n}(1,1)$, where $n \geq 2$. Suppose its minimal resolution $\nu : S \rightarrow X$ is obtained from $\mathbb{F}_n$ by m point blow-ups whose centers avoid the negative section C_0 and its successive strict transforms, and that the exceptional locus of ν is the strict transform of C_0 . Then*

$$D^b(\mathrm{coh} X) = \langle D^b(K(n,1)\text{-mod}), F_1, \dots, F_{m+2} \rangle,$$

where all F_i are exceptional. The centers may be infinitely near, provided they lie over the smooth locus and avoid the contracted curve.

Proof. We pull back the three components of the decomposition on $\mathbb{F}_n$ in Proposition 4.5 through the successive blow-ups. At each blow-up β with exceptional curve D , Orlov's decomposition, written with the exceptional object on the right, is

$$D^b(\mathrm{coh} S') = \langle L\beta^* D^b(\mathrm{coh} S), \mathcal{O}_D \rangle.$$

By Lemma 4.4, the first component remains untwisted adherent to the strict transform of C_0 . The other two initial components and the m new components are generated by exceptional objects. The new exceptional objects are understood via derived pullback from their creation stages, as in Lemma 4.4; full faithfulness preserves exceptionality. The decomposition is compatible with ν , whose only exceptional curve is assigned to the first component. By Theorem 4.2, all other components descend equivalently, and [23, Theorem 3.16] identifies the first with $D^b(K(n,1)\text{-mod})$. $\square$

For the one-point Cavey–Prince cascades from $\mathbb{P}(1,1,n)$ [10], the corollary gives decompositions whenever the minimal resolutions admit blow-up presentations satisfying its hypotheses. For $n = 3$, the KKS decomposition of $D^b(\mathrm{coh} \mathbb{P}(1,1,3))$ is $\langle D^b(K(3,1)\text{-mod}), \mathcal{O}, \mathcal{O}(3) \rangle$, as used in Section 5.

For several singularities, constructing simultaneous subcategories adherent to the exceptional curves is a separate problem. KKS gives a global obstruction in terms of the divisor class group.

Theorem 4.7 (KKS obstruction, [23, Corollary 4.8]). *Let X be a normal projective rational surface with cyclic quotient singularities, and let $\tau : Y \rightarrow X$ be its minimal resolution with exceptional divisor $\mathcal{D}$. Suppose there is a semiorthogonal decomposition*

$$D^b(\mathrm{coh} Y) = \langle \tilde{\mathcal{A}}_1, \dots, \tilde{\mathcal{A}}_t \rangle, \quad \mathcal{D} = \bigsqcup_{i=1}^t \mathcal{D}_i,$$

such that every nonempty $\mathcal{D}_i$ is a connected component of $\mathcal{D}$ and every $\tilde{\mathcal{A}}_i$ is untwisted adherent to $\mathcal{D}_i$ in the sense of [23, Definition 3.6]. For $\mathcal{D}_i = \emptyset$, adherence means that $\tilde{\mathcal{A}}_i$ is generated by one line bundle, using the convention of [23, Remark 3.8]. Then $\mathrm{Cl}(X)$ is torsion-free.

Proof. KKS Corollary 3.18 gives a decomposition into derived categories of local finite-dimensional algebras and explicitly allows empty chains; for an empty chain the algebra is $\mathbb{C}$. Additivity makes $G_0(X)$ free, and $G_0(X)_{\mathrm{tors}} \cong \mathrm{Cl}(X)_{\mathrm{tors}}$. This is the proof of [23, Corollary 4.8], using [23, Corollary 3.18, Lemma 4.1, and Proposition 4.4]. $\square$

Bright's exact sequence translates this obstruction into a Brauer-group calculation. We use the cohomological Brauer group $\mathrm{Br}(X) = H_{\mathrm{\acute{e}t}}^2(X, \mathbb{G}_m)_{\mathrm{tors}}$. In the present projective setting, its torsion classes are represented by Azumaya algebras by Gabber's theorem, as used in [9, proof of Theorem 6]; thus no distinction between the two Brauer groups arises for the finite groups computed here. This is also the Brauer group entering the obstruction in [23, Corollary 4.8].

Lemma 4.8 (Bright's exact sequence [9, Proposition 1 and Section 3]). *Let X be a normal projective surface with rational singularities and minimal resolution $\tau : Y \rightarrow X$, where Y is a rational surface. Let $\Lambda_\tau \subset \mathrm{Pic}(Y)$ denote the subgroup generated by the exceptional curves. The intersection pairing induces a homomorphism*

$$\theta : \mathrm{Pic}(Y) \longrightarrow \Lambda_\tau^\vee, \quad D \longmapsto (E \mapsto D \cdot E),$$

and there is an exact sequence

$$0 \longrightarrow \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(Y) \xrightarrow{\theta} \Lambda_\tau^\vee \longrightarrow \mathrm{Br}(X) \longrightarrow \mathrm{Br}(Y).$$

Here $\mathrm{Br}(Y) = 0$.

Corollary 4.9. *Let X be a normal projective surface with quotient singularities and minimal resolution $\tau : Y \rightarrow X$, where Y is a rational surface. For the intersection map of Lemma 4.8, there is an isomorphism*

$$\mathrm{Br}(X) \cong \mathrm{coker}(\theta : \mathrm{Pic}(Y) \longrightarrow \Lambda_\tau^\vee).$$

In particular, $\mathrm{Br}(X) = 0$ if and only if there are divisor classes $L_i \in \mathrm{Pic}(Y)$ with $L_i \cdot E_j = \delta_{ij}$ for all exceptional curves E_j .

Proof. Quotient singularities are rational, so Lemma 4.8 applies, and $\mathrm{Br}(Y) = 0$. Hence $\mathrm{Br}(X) \cong \mathrm{coker}(\theta)$. Therefore $\mathrm{Br}(X) = 0$ exactly when θ is surjective, equivalently when every dual basis vector is represented by a divisor class L_i , that is, $L_i \cdot E_j = \delta_{ij}$. $\square$

All log del Pezzo surfaces in this paper satisfy the rationality hypothesis by Proposition 2.2. In this setting, the vanishing in Corollary 4.9 is equivalent to the KKS torsion-free condition: the Weil divisor class group $\mathrm{Cl}(X)$ has no torsion [23, Definition 4.7]. It is necessary for an untwisted adherent decomposition by [23, Corollary 4.8].

The rigid base of the one-point Corti–Heuberger cascade is $\mathbb{P}(1, 1, 3)$. The exceptional lattice of $\mathbb{F}_3 \rightarrow \mathbb{P}(1, 1, 3)$ is $\mathbb{Z}C_0$. The relation $f \cdot C_0 = 1$ makes the intersection map surjective, so Corollary 4.9 gives

$$\mathrm{Br}(\mathbb{P}(1, 1, 3)) = 0.$$

The corresponding KKS decomposition of $\mathbb{P}(1, 1, 3)$ is given by Proposition 4.5 with $n = 3$. Appendix B computes the Brauer obstruction throughout the six Corti–Heuberger cascades.

5 Corti–Heuberger cascades and the surface X_{10}

We now specialize the preceding constructions to log del Pezzo surfaces with only $\frac{1}{3}(1, 1)$ -singularities. The Corti–Heuberger classification organizes the Fano index one families into six cascades from rigid bases; after recalling the deformation terminology, we construct the corresponding exceptional collections and then treat X_{10} explicitly.

Definition 5.1 ($\mathbb{Q}$ -Gorenstein deformations and rigidity). Let X be a normal projective surface with quotient singularities. A $\mathbb{Q}$ -Gorenstein deformation of X is a flat deformation that locally lifts to equivariant deformations of the index-one, or canonical, covers of the quotient singularities, namely the local cyclic covers on which the canonical divisor becomes Cartier. For projective surfaces we use proper families, and $\mathbb{Q}$ -Gorenstein deformation equivalence means being connected by a chain of such families over connected bases [2, Basic Concepts, Definition 2 and proof of Lemma 6]. A surface is $\mathbb{Q}$ -Gorenstein rigid if every sufficiently small $\mathbb{Q}$ -Gorenstein deformation is trivial.

Rigidity of the surface is a global condition, distinct from rigidity of its singularities. The singularity $\frac{1}{3}(1, 1)$ is locally $\mathbb{Q}$ -Gorenstein rigid, so small $\mathbb{Q}$ -Gorenstein deformations preserve its analytic type; the degree is also locally constant [11, Section 1.1.1].

Corti–Heuberger classify the non-smooth del Pezzo surfaces with only $\frac{1}{3}(1, 1)$ -points into 29 families, grouped into six cascades [11, Theorems 3–4]. The operations relating these families are described by the following terminology.

Definition 5.2 (Floating (-1) -curves and cascades). Let X be a log del Pezzo surface with only $\frac{1}{3}(1, 1)$ -points. A *floating (-1) -curve* on X is a smooth rational curve contained in its smooth locus with self-intersection -1 [11, Definition 5]. A *cascade* connects the deformation families by blow-ups at smooth points, or equivalently by reversing contractions of floating (-1) -curves.

Contracting a floating (-1) -curve is the inverse of a blow-up at a smooth point and increases K^2 by one. These operations preserve the singular points and their analytic types.

Definition 5.3 (Marked blow-up presentations). A *marked blow-up presentation* of a smooth surface Y is a sequence of point blow-ups

$$Y_0 \xleftarrow{\beta_1} Y_1 \xleftarrow{\beta_2} \dots \xleftarrow{\beta_m} Y_m \simeq Y,$$

with fixed identifications of the starting and final surfaces and, at each stage, a specified point $q_i \in Y_{i-1}$, its incidences with the strict transforms of previously introduced exceptional curves, and a label for the new exceptional divisor $D_i \subset Y_i$. For a marked blow-up presentation, the centers may be infinitely near points of Y_0 . When Y is a resolution of a surface with quotient singularities, the marking also specifies the curves contracted to those singularities. A *marked cascade presentation* specifies distinct smooth points on the fixed surface Z_k , their order, and the identification of their blow-up with the member X .

The distinctness in a cascade follows from ampleness. In any sequence of blow-ups at smooth points ending in a log del Pezzo surface X , an exceptional curve has anticanonical degree one when created. If $h \geq 1$ later centers lie on its successive strict transforms, its final strict transform D , still in the smooth locus, satisfies

$$(-K_X) \cdot D = 1 - h \leq 0, \tag{3}$$

contrary to ampleness. Thus the cascade centers are distinct on the base, and their exceptional curves are disjoint. This restriction does not apply to the blow-ups forming the minimal resolutions in Appendix C.

We follow [11, Theorem 4]: $X_{k,d}$ denotes the Fano index one $\mathbb{Q}$ -Gorenstein deformation family with k singular points and $d = K_X^2$. By [11, Theorem 6(1)–(6) and Remark 7], the

six rigid bases are unique in their respective base families, apart from the two auxiliary families discussed below; we denote them by Z_k . Table 1 collects their invariants and the numerical data of the marked blow-up presentations constructed in Appendix C. Here Y_0 is the starting smooth surface, $N_k = \text{rk } K_0(\tilde{Z}_k)$, and $\ell_k = N_k + k$ is the length of the resulting full exceptional collection on the canonical stack. We use $S_{1,25/3}$ for $\mathbb{P}(1, 1, 3)$, of Fano index five, to distinguish it from the Fano index one notation $X_{k,d}$.

k	Z_k	$K_{Z_k}^2$	Y_0	Number of blow-ups	N_k	ℓ_k
1	$S_{1,25/3} = \mathbb{P}(1, 1, 3)$	$25/3$	$\mathbb{F}_3$	0	4	5
2	$X_{2,17/3}$	$17/3$	$\mathbb{F}_3$	3	7	9
3	$X_{3,5}$	5	$\mathbb{F}_3$	4	8	11
4	$X_{4,7/3}$	$7/3$	$\mathbb{F}_2$	7	11	15
5	$X_{5,5/3}$	$5/3$	$\mathbb{P}^2$	9	12	17
6	$X_{6,2}$	2	$\mathbb{F}_0$	8	12	18

Table 1: Rigid cascade bases, marked blow-up presentations, and collection lengths.

Proposition 5.4. *Each minimal resolution $\tilde{Z}_k$ has a marked blow-up presentation by successive point blow-ups of a smooth surface Y_0 , with infinitely near centers allowed. The starting surfaces and numbers of blow-ups are specified in Table 1. Appendix C records the incidence data, lists the centers, and constructs the resulting Orlov full exceptional collections. Applying the Ishii–Ueda embedding and adjoining the residual-gerbe objects gives full exceptional collections on the canonical stacks of the bases, of lengths ℓ_k as recorded in Table 1 and given by Corollary 3.7.*

By [11, Corollary 8(1)–(6)], every surface in $X_{k,d}$ is obtained from Z_k by $K_{Z_k}^2 - d$ blow-ups at smooth points, distinct by (3). Orlov’s formula extends the base collection through these blow-ups, and the Ishii–Ueda embedding, together with the residual-gerbe objects, gives the collection on the canonical stack in the order of Corollary C.3. Invariance of the Brauer group under smooth blow-ups (Lemma B.5) reduces its computation throughout each cascade to the six bases treated in Appendix B.

The two auxiliary Fano-index-two families $B_{1,16/3}$ and $B_{2,8/3}$ are treated separately in Appendix B.5, using the alternative contraction sequences of [11, Remark 9].

The collection length is given by Corollary 3.7. We single out X_{10} because its hypersurface description can be compared directly with its presentation as an eight-point blow-up of $\mathbb{P}(1, 1, 3)$; this comparison underlies the parameter-space result of Appendix A and makes the exceptional collection and KKS decomposition explicit. Consider a degree-10 hypersurface $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$, writing x_0, x_1, x_2, x_3 for coordinates of weights 1, 2, 3, 5. The singularities of a general member are standard [22, 20]; we verify them for an arbitrary quasismooth member.

Lemma 5.5. *A general degree-10 hypersurface in $\mathbb{P}(1, 2, 3, 5)$ is quasismooth. Every quasismooth member avoids the coordinate points of weights 2 and 5 and has a unique singular point $p = [0 : 0 : 1 : 0]$, of type $\frac{1}{3}(1, 1)$.*

Proof. For any quasismooth member, the coefficients of x_1^5 and x_3^2 are nonzero: these are the only degree-ten monomials whose first derivatives can be nonzero on the respective punctured coordinate axes. Otherwise the corresponding axis would lie in the cone and be singular. Thus the hypersurface avoids the ambient singular points of weights 2 and 5.

At p , the quotient chart has weights $(1, 2, 5) \equiv (1, 2, 2) \pmod{3}$. Quasismoothness forces a nonzero coefficient of $x_0 x_2^3$, the only monomial giving a linear term there. Eliminating x_0 leaves tangent weights $(2, 2)$, equivalent to $(1, 1)$. These are all the ambient singular points, so quasismoothness implies smoothness elsewhere. $\square$

For every member satisfying [Lemma 5.5](#), weighted adjunction and the intersection formula of [\[14\]](#) give

$$\omega_{X_{10}}^\vee = \mathcal{O}_{X_{10}}(1), \quad K_{X_{10}}^2 = \frac{1}{3}.$$

The anticanonical class is ample, so X_{10} is a log del Pezzo surface. The higher Fano index families in [\[11, Theorem 3\]](#) have different numerical invariants, so the Fano index is one; [\[11, Theorem 4 and Table 2\]](#) identifies the family as $X_{1,1/3}$.

By [\[11, Corollary 8\(1\)\]](#), every such X_{10} is the blow-up of $\mathbb{P}(1, 1, 3)$ at eight smooth points, distinct by [\(3\)](#). Its minimal resolution is therefore the blow-up of $\mathbb{F}_3$ at eight distinct points off C_0 . The following lemma and proposition relate general eight-point configurations to general hypersurfaces. The ring presentation is due to Reid–Suzuki [\[30, Section 1 and Table 1.1\]](#).

Lemma 5.6 (Reid–Suzuki). *For eight distinct general smooth points $P_1, \dots, P_8 \in \mathbb{P}(1, 1, 3)$, the blow-up $S = \text{Bl}_{P_1, \dots, P_8} \mathbb{P}(1, 1, 3)$ has ample anticanonical divisor and*

$$R(S, -K_S) \cong \frac{\mathbb{C}[x, y, z, w]}{(f_{10})}, \quad \deg(x, y, z, w) = (1, 2, 3, 5),$$

where f_{10} is weighted homogeneous of degree 10. Consequently, S is isomorphic to a quasismooth hypersurface $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$.

Proof. By [\[11, Corollary 8\(1\)\]](#), an ample member is obtained by blowing up eight distinct smooth points of $\mathbb{P}(1, 1, 3)$. The parameter space of ordered distinct smooth eight-tuples is irreducible, and ampleness of $-3K$ is open in the corresponding family of blow-ups [\[32, Tag 0D2S\]](#). Thus $-K_S$ is ample for general centers. Reid and Suzuki compute its anticanonical ring in the displayed form [\[30, Section 1 and Table 1.1\]](#); ampleness then gives $S \simeq \text{Proj } R(S, -K_S)$.

The resulting hypersurface is quasismooth. Its punctured anticanonical cone is locally a $\mathbb{G}_m$ -bundle over the smooth locus of S . Over its $\frac{1}{3}(1, 1)$ -point it is locally the quotient of $\mathbb{A}^2 \times \mathbb{G}_m$ by μ_3 , whose faithful action on the second factor is free. Hence the punctured cone is smooth. $\square$

For a general hypersurface, the following proposition strengthens the eight-point presentation by allowing the centers to be general.

Proposition 5.7 (The general hypersurface). *A general quasismooth $X_{10} \subset \mathbb{P}(1, 2, 3, 5)$ is isomorphic to the anticanonical model of the blow-up of $\mathbb{P}(1, 1, 3)$ at eight distinct general smooth points. Its minimal resolution is therefore the blow-up of $\mathbb{F}_3$ at eight distinct points away from the negative section C_0 .*

Appendix [A](#) compares the parameter spaces of eight-point configurations and degree-10 hypersurfaces. Parameter counts and finiteness of blow-up presentations show that the resulting hypersurfaces form a dominant family.

The exceptional collection and its Euler matrix can be computed from the eight-point blow-up presentation. For any X_{10} satisfying the hypotheses of [Theorem 1.3](#), let

$$\sigma : \tilde{X}_{10} \longrightarrow \mathbb{F}_3$$

be a marked blow-up presentation at eight distinct points off C_0 . Write $D_1, \dots, D_8$ for its pairwise disjoint exceptional curves and f for the fiber class on $\mathbb{F}_3$. Let

$$\nu : \tilde{X}_{10} \longrightarrow X_{10}$$

be the minimal resolution, and let E be the strict transform of C_0 . The computation at the end of [Section 4](#), together with invariance under smooth blow-ups ([Lemma B.5](#)), gives $\mathrm{Br}(X_{10}) = 0$.

Orlov's formula extends the standard collection on $\mathbb{F}_3$ through the eight blow-ups. Applying the Ishii–Ueda embedding and adjoining the residual-gerbe object gives the collection on the canonical stack, of length 13 by [Corollary 3.7](#), displayed in [Theorem 1.3](#). The curves D_i are pairwise disjoint, so their exceptional objects may be listed in any order.

Corollary 5.8. *Let $\pi : \mathcal{X}_{10} \rightarrow X_{10}$ be the canonical stack. The category $D^b(\mathrm{coh} \mathcal{X}_{10})$ has a full exceptional collection of length 13.*

More generally, rename the parameter in the Hille–Perling family (1) as t and set $t = s + 4$ [[19](#), Proposition 5.2]. This gives the following collections; $s = -1$ is the standard one.

Proposition 5.9 (The Euler matrix for $\mathcal{X}_{10}$). *Let $s \in \mathbb{Z}$ and let*

$$\begin{aligned} \mathcal{C}(s) = \langle & \mathcal{E}_p, \Phi(\mathcal{O}_{D_1}(-1)), \dots, \Phi(\mathcal{O}_{D_8}(-1)), \Phi(\sigma^*\mathcal{O}), \Phi(\sigma^*\mathcal{O}(f)), \\ & \Phi(\sigma^*\mathcal{O}(C_0 + (s+4)f)), \Phi(\sigma^*\mathcal{O}(C_0 + (s+5)f)) \rangle \end{aligned}$$

be an exceptional collection of $D^b(\mathrm{coh} \mathcal{X}_{10})$ associated with the Hille–Perling toric system for $\mathbb{F}_3$. The Euler matrix $S_{ij}(s) = \chi(\mathcal{C}_i(s), \mathcal{C}_j(s))$ is

$$S_{\mathcal{X}_{10}}(s) = \left(\begin{array}{c|c|cccc} 1 & 0_{1 \times 8} & 0 & -1 & -(s+1) & -(s+2) \\ \hline 0_{8 \times 1} & I_8 & \hline & & -\mathbf{1}_{8 \times 4} \\ \hline & & 1 & 2 & 2s+7 & 2s+9 \\ & & 0 & 1 & 2s+5 & 2s+7 \\ & & 0 & 0 & 1 & 2 \\ & & 0 & 0 & 0 & 1 \end{array} \right).$$

Proof. This is the specialization $n = 3$, $m = 8$ of [Proposition 6.3](#). The curves D_i are disjoint, so their objects may be ordered arbitrarily. The four line bundles have degrees $0, 1, s+1, s+2$ on E , giving the displayed first row. $\square$

For the singular surface X_{10} , we extend the KKS decomposition of $\mathbb{P}(1, 1, 3)$ recalled in [Section 4](#) through the eight blow-ups; [Lemma 4.4](#) preserves the adherent component.

Theorem 5.10. *Every X_{10} satisfying the hypotheses of [Theorem 1.3](#) admits a semiorthogonal decomposition*

$$D^b(\mathrm{coh} X_{10}) = \left\langle D^b(K(3, 1)\text{-mod}), F_1, \dots, F_{10} \right\rangle,$$

where $\eta : X_{10} \rightarrow \mathbb{P}(1, 1, 3)$ is the blow-down of the chosen presentation, ν is the minimal resolution, and

$$\begin{aligned} F_1 &= \mathcal{O}_{X_{10}}, & F_2 &= \eta^*\mathcal{O}_{\mathbb{P}(1,1,3)}(3), \\ F_{i+2} &= \mathcal{O}_{\nu(D_i)} & (1 \leq i \leq 8) \end{aligned}$$

are exceptional objects.

Proof. [Corollary 4.6](#), with $n = 3$ and $m = 8$, gives this decomposition using the eight-point presentation valid for every quasismooth X_{10} . Since the centers are distinct and ν is an isomorphism near every D_i , the objects $\mathcal{O}_{D_i}$ in Orlov's decomposition descend to $\mathcal{O}_{\nu(D_i)}$. If $\tau : \mathbb{F}_3 \rightarrow \mathbb{P}(1, 1, 3)$ contracts C_0 , then $\tau\sigma = \eta\nu$ and $\tau^*\mathcal{O}_{\mathbb{P}(1,1,3)}(3) = \mathcal{O}_{\mathbb{F}_3}(C_0 + 3f)$. The projection formula and $R\nu_*\mathcal{O} = \mathcal{O}_{X_{10}}$ therefore identify the two descended line bundles with F_1, F_2 . $\square$

Corollary 5.11. *Let X belong to a Fano index one family $X_{1,d}$. Choose a marked cascade presentation by blowing up m distinct smooth points of $\mathbb{P}(1, 1, 3)$, so that $m = 25/3 - d$. Then*

$$D^b(\text{coh } X) = \langle D^b(K(3, 1)\text{-mod}), F_1, \dots, F_{m+2} \rangle,$$

where $F_1, \dots, F_{m+2}$ are exceptional objects.

Proof. The marked cascade presentation induces the marked blow-up presentation required by [Corollary 4.6](#), with $n = 3$; its centers are distinct on $\mathbb{F}_3$ and avoid the negative section by (3). Apply that corollary. $\square$

6 Euler matrices for exceptional collections on canonical stacks

[Corollary 3.6](#) supplies the full exceptional collections on the canonical stacks considered here. We give a block formula for their Euler matrices, separating the local contributions from the Euler matrix of the resolution collection and the pairings between them. For a line-bundle collection, these pairings reduce to intersection numbers with the exceptional curves. We then specialize to a single cyclic point resolved by blow-ups of a Hirzebruch surface and compute every block explicitly.

Proposition 6.1 (Mixed-basket Euler block formula). *Let X be a normal projective rational surface with singular points $p_1, \dots, p_r$ of types $\frac{1}{n_i}(1, 1)$, where $n_i \geq 2$, and let $E_i \subset \tilde{X}$ be the exceptional curve over p_i . Fix a full exceptional collection $\langle G_1, \dots, G_N \rangle$ on the minimal resolution. For the associated full exceptional collection on the canonical stack of [Corollary 3.6](#), the Euler matrix is*

$$S = \begin{pmatrix} T_{n_1} & 0 & \cdots & 0 & B_1 \\ 0 & T_{n_2} & \cdots & 0 & B_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & T_{n_r} & B_r \\ 0 & 0 & \cdots & 0 & H \end{pmatrix}, \quad H_{ab} = \chi(G_a, G_b).$$

For an arbitrary exceptional collection on the resolution, the off-diagonal blocks B_i are defined by $(B_i)_{j,a} = \chi(\mathcal{E}_{p_i,j}, \Phi(G_a))$, where $2 \leq j \leq n_i - 1$. The local block is

$$T_n = (I_{n-2} - J_{n-2})^2, \quad (J_{n-2})_{ab} = \delta_{b,a+1} \quad (1 \leq a, b \leq n-2), \quad (4)$$

where J_{n-2} is the nilpotent shift with ones immediately above the diagonal. For $n = 2$, the block is empty.

When the resolution collection consists of line bundles $G_a = \mathcal{O}(A_a)$, these pairings are determined by intersection numbers alone. Put $d_{ia} = A_a \cdot E_i$. For $n_i \geq 4$,

$$(B_i)_{j,a} = \begin{cases} 0, & 2 \leq j \leq n_i - 3, \\ d_{ia} + 1, & j = n_i - 2, \\ -d_{ia}, & j = n_i - 1. \end{cases}$$

For $n_i = 3$, the sole row is $(B_i)_{2,a} = -d_{ia}$; for $n_i = 2$, B_i has no rows. Thus the resolution matrix H and the intersection numbers $A_a \cdot E_i$ determine the entire Euler matrix for a line-bundle collection on the resolution.

Proof. The equivariant Koszul resolution on $[\mathbb{A}^2/\mu_n]$ gives

$$\chi(\mathcal{O}_0 \otimes \rho_u, \mathcal{O}_0 \otimes \rho_v) = \delta_{uv} - 2\delta_{v,u+1} + \delta_{v,u+2}.$$

Within $2 \leq u, v \leq n-1$, no cyclic wraparound term occurs, so this is T_n . Distinct local collections have disjoint support, which gives the off-diagonal zeros among them. Full faithfulness of Φ gives

$$\chi(\Phi(G_a), \Phi(G_b)) = \chi(G_a, G_b),$$

so the lower-right block is exactly H ; [Theorem 3.4](#) supplies the lower-left zero blocks.

Let Ψ be the left adjoint of Φ . Left adjunction gives

$$\mathrm{RHom}_{\mathcal{X}}(\mathcal{E}_{p_i,j}, \Phi(G_a)) \simeq \mathrm{RHom}_{\tilde{X}}(\Psi(\mathcal{E}_{p_i,j}), G_a).$$

[Lemma 3.8](#), equivalently (2), computes $\Psi(\mathcal{E}_{p_i,j})$ locally near E_i . For the indices belonging to the non-special collection, it gives

$$\Psi(\mathcal{E}_{p_i,n_i-2}) = \mathcal{O}_{E_i}(-n_i)[1], \quad \Psi(\mathcal{E}_{p_i,n_i-1}) = \mathcal{O}_{E_i}(-n_i+1),$$

with all earlier non-special characters sent to zero. The first displayed source belongs to the local collection only when $n_i \geq 4$, and the second only when $n_i \geq 3$. For the divisor inclusion $E_i \hookrightarrow \tilde{X}$, duality and $E_i^2 = -n_i$ give

$$\chi_{\tilde{X}}(\mathcal{O}_{E_i}(t), \mathcal{O}(A_a)) = -\chi_{\mathbb{P}^1}(\mathcal{O}(d_{ia} - n_i - t)).$$

Together with $\chi(\mathcal{O}_{\mathbb{P}^1}(q)) = q+1$, this gives $d_{ia}+1$ for the shifted penultimate image and $-d_{ia}$ for the final image. $\square$

Remark 6.2 (Local checks and orthogonality). For a vector bundle V on $[\mathbb{A}^2/\mu_n]$, gerbe adjunction and the ring-action convention of [Section 3](#) give

$$\mathrm{RHom}(\mathcal{O}_0 \otimes \rho_j, V) \simeq (V_0 \otimes \rho_{-j-2})^{\mu_n}[-2].$$

Its Euler characteristic is the multiplicity of ρ_{j+2} in the fiber V_0 . In particular, $\chi(\mathcal{E}_j, \mathcal{L}_a) = \delta_{a,j+2}$, with indices read modulo n .

We also use the images of the special local generators. For the special characters $j = 0, 1$, the corresponding free sheaves belong to the image of ϕ [[21](#), Proposition 1.1], so $\phi(P_0) = \mathcal{L}_0$ and $\phi(P_{-1}) = \mathcal{L}_1$. Applying ϕ to the Euler sequence $0 \rightarrow P_{-1} \rightarrow P_0^{\oplus 2} \rightarrow P_1 \rightarrow 0$ yields

$$\phi(P_1) \simeq \mathrm{coker}(\mathcal{L}_1 \xrightarrow{(-y,x)} \mathcal{L}_0^{\oplus 2}) \simeq (x, y) \otimes \rho_{n-1}.$$

The coordinate maps are those of the universal cluster. Thus, for $n \geq 3$,

$$0 \longrightarrow \phi(P_1) \longrightarrow \mathcal{L}_{n-1} \longrightarrow \mathcal{E}_{n-1} \longrightarrow 0.$$

Hence $\chi(\mathcal{E}_{n-1}, \phi(P_1)) = -1$. This verifies the Euler pairing with the image of $\mathcal{O}(f)$, including the -1 in the X_{10} Euler matrix. The fiber-multiplicity rule cannot be applied to $\phi(P_1)$ itself: it is torsion-free but not locally free at the origin.

For $n \geq 5$, the objects $\mathcal{E}_j$ with $2 \leq j \leq n-3$ are orthogonal to the resolution image on both sides: one direction is semiorthogonality, and the other follows from $\psi(\mathcal{E}_j) = 0$. This does not split off an orthogonal summand of the whole category, since $\chi(\mathcal{E}_j, \mathcal{E}_{j+1}) = -2$ and $\chi(\mathcal{E}_j, \mathcal{E}_{j+2}) = 1$.

The local block depends only on the singularity type, while the resolution contribution depends on the chosen collection and, for line bundles, only on its intersection numbers with the exceptional curves. We now specialize [Proposition 6.1](#) to a single $\frac{1}{n}(1, 1)$ -point whose resolution is obtained from $\mathbb{F}_n$ by m point blow-ups away from the successive strict transforms of C_0 .

To describe the collection and its matrix blocks, let $n \geq 2$ and $m \geq 0$. Suppose that a normal projective surface X has a minimal resolution fitting into a diagram

$$\tilde{X} \xrightarrow{\sigma} \mathbb{F}_n, \quad \tilde{X} \xrightarrow{\nu} X,$$

where σ is a sequence of m point blow-ups, possibly infinitely near, whose centers avoid the successive strict transforms of the negative section $C_0 \subset \mathbb{F}_n$, and ν contracts the strict transform E of C_0 to the unique singular point of X , of type $\frac{1}{n}(1, 1)$. Let D_a be the exceptional curve on the surface produced by the a -th blow-up and G_a the derived pullback of $\mathcal{O}_{D_a}(-1)$ from that surface to $\tilde{X}$. We use the intersection conventions

$$C_0^2 = -n, \quad C_0 \cdot f = 1, \quad f^2 = 0.$$

Write $\mathcal{E}_i = \mathcal{E}_{p,i}$ for the objects of [Definition 3.3](#) at the unique singular point, and retain the notation $\mathcal{X}, \Phi$ of [Proposition 6.1](#). For $s \in \mathbb{Z}$, the projective-bundle collection (1), with its parameter renamed t and set to $t = s + n + 1$, together with Orlov's blow-up formula and the Ishii–Ueda decomposition, gives the ordered full exceptional collection

$$\begin{aligned} \mathcal{C}_{n,m}(s) = \langle & \mathcal{E}_2, \dots, \mathcal{E}_{n-1}, \Phi(G_m), \dots, \Phi(G_1), \\ & \Phi(\sigma^*\mathcal{O}), \Phi(\sigma^*\mathcal{O}(f)), \Phi(\sigma^*\mathcal{O}(C_0 + (s + n + 1)f)), \\ & \Phi(\sigma^*\mathcal{O}(C_0 + (s + n + 2)f)) \rangle. \end{aligned}$$

The local block is T_n from (4). For $n \geq 4$, the local-to-Hirzebruch block $B_n(s) \in \text{Mat}_{(n-2) \times 4}(\mathbb{Z})$ is

$$B_n(s) = \begin{pmatrix} & & 0_{(n-4) \times 4} & \\ 1 & 2 & s+2 & s+3 \\ 0 & -1 & -(s+1) & -(s+2) \end{pmatrix}, \quad (5)$$

where the zero block is absent for $n = 4$. The penultimate and final rows are indexed by $\mathcal{E}_{n-2}$ and $\mathcal{E}_{n-1}$, respectively. For $n = 3$, the local collection consists only of $\mathcal{E}_2$, so $T_3 = (1)$. The representation $\rho_{n-2} = \rho_1$ is special and does not belong to the Ishii–Ueda complement; consequently, only the final row of $B_n(s)$ occurs:

$$B_3(s) = \begin{pmatrix} 0 & -1 & -(s+1) & -(s+2) \end{pmatrix}.$$

For $n = 2$, both T_2 and $B_2(s)$ are empty, and the collection consists only of the resolution objects.

The Euler matrix $H_n(s) \in M_4(\mathbb{Z})$ of the four line bundles on $\mathbb{F}_n$ is

$$H_n(s) = \begin{pmatrix} 1 & 2 & 2s+n+4 & 2s+n+6 \\ 0 & 1 & 2s+n+2 & 2s+n+4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Proposition 6.3 (The one-point specialization for a Hirzebruch surface). *For the collection $\mathcal{C}_{n,m}(s)$ and the blocks defined above, the $(n+m+2) \times (n+m+2)$ matrix of the Euler form is*

$$S_{n,m}(s) = \begin{pmatrix} T_n & 0 & B_n(s) \\ 0 & I_m & -\mathbf{1}_{m \times 4} \\ 0 & 0 & H_n(s) \end{pmatrix}.$$

Here $\mathbf{1}_{m \times 4}$ denotes the matrix all of whose entries are equal to 1.

Proof. Proposition 6.1 gives T_n and the local pairings with the line bundles. Since σ is an isomorphism near E , $(\sigma^*M) \cdot E = M \cdot C_0$ for a divisor class M on $\mathbb{F}_n$. For the four line bundles in $\mathcal{C}_{n,m}(s)$, the intersection numbers are

$$M \cdot C_0 = 0, \quad 1, \quad s+1, \quad s+2.$$

These values give (5), including the stated truncation when $n = 3$. The adjoint images in Lemma 3.8 are supported on E . Since every G_a is supported away from E , the remaining local pairings satisfy

$$\chi(\mathcal{E}_i, \Phi(G_a)) = 0$$

for every i and a .

For a blow-up $\beta : S' \rightarrow S$ at p , with exceptional curve D , divisor duality gives, for any perfect complex A , an isomorphism of complexes of vector spaces and the resulting value of the Euler form:

$$\mathrm{RHom}_{S'}(\mathcal{O}_D(-1), L\beta^*A) \simeq Li_p^*A[-1], \quad \chi(\mathcal{O}_D(-1), L\beta^*A) = -\mathrm{rk}(A).$$

Earlier Orlov objects have rank zero, and line bundles have rank one. Induction, full faithfulness, and Orlov semiorthogonality [29, Theorem 4.3 and Corollary 4.4] therefore give

$$\chi(G_a, G_b) = \delta_{ab}, \quad \chi(G_a, \sigma^*L) = -1, \quad \chi(\sigma^*L, G_a) = 0$$

for every line bundle L on $\mathbb{F}_n$. These pairings yield I_m , $-\mathbf{1}_{m \times 4}$, and the opposite zero block, even for infinitely near centers. In that case the off-diagonal Ext groups need not vanish, but their Euler characteristics still do.

The block $H_n(s)$ follows from Riemann–Roch on $\mathbb{F}_n$. Using

$$K_{\mathbb{F}_n} = -2C_0 - (n+2)f$$

together with

$$\chi(\mathcal{O}(A), \mathcal{O}(B)) = 1 + \frac{1}{2}(B-A) \cdot (B-A-K_{\mathbb{F}_n}),$$

gives $H_n(s)$ for the four Hille–Perling line bundles and completes the specialization of Proposition 6.1. $\square$

For $n = 3$ and $m = 8$, one has $T_3 = (1)$ and $B_3(s) = (0, -1, -(s+1), -(s+2))$, while $H_3(s)$ is the 4×4 block in Proposition 5.9. Together with I_8 and $-\mathbf{1}_{8 \times 4}$, the formula for $S_{n,m}(s)$ recovers $S_{\mathcal{X}_{10}}(s)$.

Proposition 6.3 also applies to the one-point Cavey–Prince cascades starting from $\mathbb{P}(1, 1, n)$ whenever their marked blow-up presentations satisfy the stated hypotheses [10], including infinitely near centers. The Orlov objects are derived pullbacks from the stages at which the corresponding exceptional divisors are created.

Concluding remarks

Semiorthogonal decompositions of $D^b(\mathrm{coh} X)$ are obtained for the one- and three-point cascades by [Corollaries 5.11](#) and [B.6](#). For the six-point cascade, [Corollary B.8](#) gives a decomposition of $D^b(\mathrm{coh}(X, \alpha))$ for a nonzero $\alpha \in \mathrm{Br}(X)$; by [Theorem 4.7](#), no untwisted adherent decomposition of the KKS type considered here exists. For $k = 2, 4, 5$, the bases Z_k are non-toric ([Remark B.7](#)), so the toric construction used above is unavailable; despite Brauer vanishing, constructing simultaneous untwisted adherent subcategories remains open. [Appendix A](#) establishes, for X_{10} , the correspondence between general degree-10 hypersurfaces and general eight-point configurations on $\mathbb{P}(1, 1, 3)$; analogous comparisons for other families require separate classifying maps and fiber analyses ([Remark C.4](#)).

For other mixed baskets, [Proposition 6.1](#) reduces the Euler calculation for an exceptional collection of line bundles on the resolution to its resolution matrix and intersection data. Two natural testing grounds are the Cavey–Prince models $X_{n_1+n_2} \subset \mathbb{P}(1, 1, n_1, n_2)$ [[10](#), [Corollary 7.1](#)] and Miura’s classification [[26](#)] of del Pezzo surfaces with $\frac{1}{3}(1, 1)$ - and $\frac{1}{4}(1, 1)$ -singularities and no floating (-1) -curves. A further question, in the spirit of the mutation-equivalence perspective recalled in the Introduction, is whether exceptional collections obtained from different marked blow-up presentations of the same surface are related by mutations. More broadly, extending these categorical constructions to wider singularity baskets would provide further B -model data for the mirror-symmetric classification program beyond the $\frac{1}{3}(1, 1)$ case.

A The general degree-ten hypersurface as a blow-up

[Section 5](#) proves the blow-up presentation for every individual hypersurface under consideration. Here we prove [Proposition 5.7](#) by comparing the parameter space of ordered eight-point configurations on $\mathbb{P}(1, 1, 3)$ with the parameter space of quasismooth degree-10 hypersurfaces in $\mathbb{P}(1, 2, 3, 5)$. The construction of [Lemma 5.6](#) associates such a hypersurface to eight general smooth points.

Throughout this appendix, put

$$\begin{aligned} U &\subset \mathbb{P}H^0(\mathbb{P}(1, 2, 3, 5), \mathcal{O}(10)), & G &= \mathrm{Aut} \mathbb{P}(1, 2, 3, 5), \\ Q &= \mathbb{P}(1, 1, 3), & H &= \mathrm{Aut} Q, \end{aligned}$$

where U is the nonempty open set of quasismooth hypersurfaces. Let W be a nonempty H -invariant open subset of the space of ordered distinct smooth eight-tuples on Q for which [Lemma 5.6](#) holds. For $P = (P_1, \dots, P_8) \in W$, write $S_P = \mathrm{Bl}_{P_1, \dots, P_8} Q$. Thus $-K_{S_P}$ is ample, $K_{S_P}^2 = 1/3$, and S_P is its own anticanonical model.

We first record the elementary dimensions of these parameter spaces. In coordinates of weights $1, 2, 3, 5$, a degree-10 monomial $x^a y^b z^c w^e$ satisfies $a + 2b + 3c + 5e = 10$. For fixed e, c , there are $\lfloor (10 - 5e - 3c)/2 \rfloor + 1$ choices of b , with a then determined. The contributions for $e = 0, 1, 2$ are $14, 5, 1$, so there are 20 monomials and U is a nonempty open subset of $\mathbb{P}^{19}$. Thus $\dim U = 19$. The dimensions of the graded pieces in degrees $1, 2, 3, 5$ are $1, 2, 3, 6$, respectively. Since invertibility is an open condition and graded coordinate changes are taken modulo the one-dimensional subgroup of weighted scalars [[12](#), [Theorem 4.2](#)],

$$\dim G = 1 + 2 + 3 + 6 - 1 = 11.$$

The space of ordered distinct smooth eight-tuples is a nonempty open subset of $(Q_{\text{sm}})^8$. Since Q_{sm} is an irreducible surface, W is irreducible of dimension 16. Finally, a graded automorphism of Q has the form

$$(u, v) \mapsto A(u, v), \quad z \mapsto az + q_3(u, v),$$

where $A \in \text{GL}_2$, $a \neq 0$, and q_3 is a binary cubic, modulo weighted scalars [12, Theorem 4.2]. Hence

$$\dim H = 4 + 1 + 4 - 1 = 8.$$

The proof now follows a dimension-count strategy. Lemma A.1 gives a general lifting result for graded hypersurface rings. Lemma A.2 identifies hypersurface isomorphism classes with G -orbits, while Lemma A.3 identifies their stabilizers with their automorphism groups. Lemma A.4 proves that the stabilizer in H of a general ordered configuration is trivial, while Lemma A.5 proves the required finiteness of blow-up presentations modulo H and of the relevant automorphism groups. Finally, Proposition A.6 constructs $F : G \times W_0 \rightarrow U$ and shows that every nonempty fiber has dimension 8. Hence

$$\dim \overline{F(G \times W_0)} = 11 + 16 - 8 = 19 = \dim U,$$

which proves dominance.

Lemma A.1 (Lifting graded hypersurface isomorphisms). *Let $S = \mathbb{C}[x_0, \dots, x_n]$ be a polynomial ring with positive grading $\deg x_i = a_i > 0$. Let $f, g \in S$ be nonzero homogeneous elements of the same degree d , and assume that $d > \max_i a_i$. Then every graded isomorphism*

$$S/(f) \simeq S/(g)$$

lifts uniquely to a graded automorphism $\Phi : S \xrightarrow{\sim} S$. Moreover, $\Phi(f) = cg$ for some $c \in \mathbb{C}^\times$.

Proof. Let $\phi : S/(f) \rightarrow S/(g)$ be a graded isomorphism. Since f and g have degree d , their ideals have no nonzero components in the generator degrees $a_i < d$. Thus the image of each x_i under ϕ has a unique homogeneous representative in S_{a_i} . Sending x_i to this representative defines a graded endomorphism $\Phi : S \rightarrow S$ inducing ϕ .

Let S_+ be the positive-degree ideal. Since $d > \max_i a_i$, every degree- d monomial has at least two positive-degree factors, and hence $f, g \in S_+^2$. Therefore the indecomposable space of each quotient ring, namely its positive-degree ideal modulo its square, identifies with S_+/S_+^2 . Consequently, ϕ induces an automorphism of S_+/S_+^2 . Let $b_1 < \dots < b_r$ be the distinct weights and let X_j denote the vector of variables of weight b_j . If a monomial of weighted degree b_j contains a variable of weight b_j , then it is that variable alone, since all weights are positive. Thus

$$\Phi(X_j) = A_j X_j + P_j(X_1, \dots, X_{j-1}),$$

where A_j represents the weight- b_j component of the induced automorphism of S_+/S_+^2 , and is therefore invertible. This is block triangular in increasing weight: after recovering $X_1, \dots, X_{j-1}$, one recovers X_j as $A_j^{-1}(\Phi(X_j) - P_j(X_1, \dots, X_{j-1}))$. Hence Φ is a graded automorphism.

Since Φ is invertible and induces ϕ , it carries (f) onto (g) . Comparing the degree- d components gives $\Phi(f) = cg$ for some $c \in \mathbb{C}^\times$. Finally, any lift of ϕ must send each generator to its unique homogeneous representative in degree $a_i < d$, so Φ is unique. $\square$

The following two lemmas apply this result to weighted hypersurfaces of Fano index one.

Lemma A.2 (Index-one hypersurface orbits). *Let $\mathbb{P} = \mathbb{P}(a_0, \dots, a_n)$, where $n \geq 2$, and let $d = \sum_{i=0}^n a_i - 1$. Two well-formed quasismooth hypersurfaces of degree d in $\mathbb{P}$ are isomorphic if and only if they lie in the same $\text{Aut}(\mathbb{P})$ -orbit.*

Proof. Write $S = \mathbb{C}[x_0, \dots, x_n]$, with $\deg x_i = a_i$, and let $X = V(f)$. Weighted adjunction [14] gives $\mathcal{O}_X(1) \simeq \mathcal{O}_X(-K_X)$ as reflexive sheaves. Hence the homogeneous coordinate ring is the intrinsic anticanonical section ring:

$$R(X, -K_X) := \bigoplus_{m \geq 0} H^0(X, \mathcal{O}_X(-mK_X)) \simeq S/(f).$$

Let $X' = V(g)$ be another such hypersurface. An isomorphism $X \simeq X'$ induces an isomorphism of their intrinsic anticanonical rings, hence a graded isomorphism $S/(f) \simeq S/(g)$. Since $d > \max_i a_i$, Lemma A.1 applies and the isomorphism lifts to a graded automorphism Φ of S satisfying $\Phi(f) = cg$ for some $c \in \mathbb{C}^\times$. The induced automorphism of $\mathbb{P}$ therefore maps X onto X' . The converse follows by restricting an ambient automorphism. $\square$

The same anticanonical-ring description identifies ambient stabilizers with intrinsic automorphism groups.

Lemma A.3 (Index-one hypersurface stabilizers). *Let $\mathbb{P}$ and d be as in Lemma A.2, and put $\Gamma = \text{Aut}(\mathbb{P})$. For every well-formed quasismooth hypersurface $X \subset \mathbb{P}$ of degree d , restriction induces an isomorphism*

$$\text{Stab}_\Gamma(X) \xrightarrow{\sim} \text{Aut}(X).$$

Proof. Write $S = \mathbb{C}[x_0, \dots, x_n]$, with $\deg x_i = a_i$, and fix $X = V(f)$. Restriction defines

$$\text{res}: \text{Stab}_\Gamma(X) \longrightarrow \text{Aut}(X).$$

We construct an inverse ι . Choose $\theta: \mathcal{O}_X(1) \xrightarrow{\sim} \omega_X^\vee$, which, as in the proof of Lemma A.2, induces a graded-ring identification $j_\theta: S/(f) \xrightarrow{\sim} R(X, -K_X)$. We will show that the resulting class in Γ is independent of this choice.

Let $\alpha \in \text{Aut}(X)$. For each $m \geq 0$, pullback along α^{-1} , followed by the natural identification of the pullback of the anticanonical sheaf, gives an automorphism

$$\alpha_m: H^0(X, \mathcal{O}_X(-mK_X)) \xrightarrow{\sim} H^0(X, \mathcal{O}_X(-mK_X)).$$

Functoriality of pullback gives $\alpha_{m+m'}(ss') = \alpha_m(s)\alpha_{m'}(s')$, so these maps define a graded-ring automorphism

$$\alpha_R: R(X, -K_X) \xrightarrow{\sim} R(X, -K_X).$$

By Lemma A.1, the transported automorphism $j_\theta^{-1}\alpha_R j_\theta$ of $S/(f)$ has a unique lift $\Phi_\alpha: S \xrightarrow{\sim} S$ preserving (f) . Set $\iota(\alpha) := [\Phi_\alpha] \in \text{Stab}_\Gamma(X)$.

To verify independence of θ , let θ' be another choice. The automorphism $\theta^{-1}\theta'$ of the rank-one reflexive sheaf $\mathcal{O}_X(1)$ is multiplication by a global unit. By [18, Chapter I, Theorem 3.4], every global regular function on the projective variety X is constant. Thus $\theta' = \lambda\theta$ for some $\lambda \in \mathbb{C}^\times$, and $j_{\theta'}$ differs from j_θ in degree m by multiplication by λ^m .

This scaling commutes with α_R , so $j_{\theta'}^{-1}\alpha_R j_{\theta'} = j_{\theta}^{-1}\alpha_R j_{\theta}$. Equivalently, the scaling on S is

$$\rho_\lambda : (x_0, \dots, x_n) \mapsto (\lambda^{a_0} x_0, \dots, \lambda^{a_n} x_n).$$

As ρ_λ belongs to the $\mathbb{G}_m$ -action defining $\mathbb{P}$, it represents the identity in Γ . Hence $[\Phi_\alpha]$ is independent of θ , so ι is well defined.

Finally, Φ_α induces α_R on the anticanonical ring. Since $X = \text{Proj } R(X, -K_X)$, its restriction to X is α , and hence $\text{res} \circ \iota = \text{id}_{\text{Aut}(X)}$. Conversely, let $\gamma \in \text{Stab}_\Gamma(X)$, choose a graded representative $\Psi \in \text{Aut}_{\text{gr}}(S)$ preserving (f) , and put $\alpha = \text{res}(\gamma)$. The quotient automorphisms induced by Ψ and $j_{\theta}^{-1}\alpha_R j_{\theta}$ define the same automorphism of $\text{Proj}(S/(f))$. Their linearizations of $\mathcal{O}_X(1)$ may differ by a scalar, so for some $\lambda \in \mathbb{C}^\times$ they differ in degree m by multiplication by λ^m . Replacing Ψ by $\rho_\lambda^{-1} \circ \Psi$ makes the quotient automorphisms equal. Uniqueness in [Lemma A.1](#) then gives $\rho_\lambda^{-1} \circ \Psi = \Phi_\alpha$. Since $[\rho_\lambda] = 1$ in Γ , we have $[\Psi] = [\Phi_\alpha]$, and $\iota \circ \text{res} = \text{id}_{\text{Stab}_\Gamma(X)}$. Therefore res is the asserted isomorphism. $\square$

Lemma A.4 (Generic configuration stabilizers). *There is a nonempty open subset of W on which the stabilizer in H of every ordered tuple is trivial.*

Proof. Let $\mathcal{C}$ be the space of ordered distinct smooth eight-tuples on Q , which is irreducible by the discussion above. Consider the open locus where all eight points lie in the chart $u \neq 0$ and at least five have distinct images in $\mathbb{P}_{[u:v]}^1$; these five points will also provide the 5×5 minor for the five residual unknowns below. An automorphism fixing the ordered tuple induces an element of PGL_2 fixing at least five distinct points, and hence the identity, since a nontrivial Möbius transformation has at most two fixed points. Thus $A = \lambda I$ for some $\lambda \in \mathbb{C}^\times$. The weighted scalar represented by λ acts as $(u, v, z) \mapsto (\lambda u, \lambda v, \lambda^3 z)$; composing with its inverse gives a representative with $A = I$. After renaming the coefficient of z and the binary cubic, the residual action in the affine coordinates $t = v/u$ and $\zeta = z/u^3$ is still $(t, \zeta) \mapsto (t, a\zeta + q(t))$, where $\deg q \leq 3$. Write $q(t) = c_0 + c_1 t + c_2 t^2 + c_3 t^3$. The condition that (t_j, ζ_j) be fixed is

$$(a - 1)\zeta_j + c_0 + c_1 t_j + c_2 t_j^2 + c_3 t_j^3 = 0, \quad 1 \leq j \leq 8.$$

These equations form a homogeneous linear system in $(a - 1, c_0, c_1, c_2, c_3)$ with coefficient matrix

$$M(P) = \begin{pmatrix} \zeta_1 & 1 & t_1 & t_1^2 & t_1^3 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \zeta_8 & 1 & t_8 & t_8^2 & t_8^3 \end{pmatrix}.$$

The rank-five condition is open, since its complement is the common zero locus of the 5×5 minors. To show that this open locus is nonempty, choose five distinct values t_j and set $\zeta_j = t_j^4$. The corresponding minor is, up to sign, the Vandermonde determinant

$$\det(t_j^{\ell-1})_{1 \leq j, \ell \leq 5} = \prod_{1 \leq i < j \leq 5} (t_j - t_i) \neq 0.$$

The chart, distinctness, and rank-five conditions therefore define a nonempty open subset $\mathcal{C}^\circ \subset \mathcal{C}$. Since $W \subset \mathcal{C}$ is also nonempty open and $\mathcal{C}$ is irreducible, $W \cap \mathcal{C}^\circ$ is nonempty open. For every tuple in this intersection, the fixing equations give $a = 1$ and $q = 0$. Thus the normalized representative is the identity, and the stabilizer in H is trivial. $\square$

The next lemma shows that a fixed surface arises from finitely many H -orbits of ordered configurations. This finiteness determines the fiber dimension of the morphism F defined below and hence the dimension of its image in U .

Lemma A.5 (Finiteness of blow-up presentations). *For $P \in W$, the surface S_P has finitely many floating (-1) -curves. A fixed isomorphism class of surfaces S_P arises from only finitely many H -orbits in W . If the stabilizer of P in H is trivial, then $\text{Aut}(S_P)$ is finite.*

Proof. Let $\nu : Y \rightarrow S_P$ be the minimal resolution and put $A = \nu^*(-K_{S_P})$. Then $A^2 = 1/3$. A floating (-1) -curve is disjoint from the singular locus, so its strict transform C on Y satisfies

$$C^2 = -1, \quad A \cdot C = 1.$$

Decompose its numerical class as $C = 3A + v$. Then

$$A \cdot v = 0, \quad v^2 = (C - 3A)^2 = C^2 - 6A \cdot C + 9A^2 = -4.$$

The Hodge index theorem makes $A^\perp$ negative definite [18, Chapter V, Theorem 1.9]. The possible vectors v lie in a translate of the numerical divisor lattice and on the compact locus $v^2 = -4$ in this negative definite space. There are therefore only finitely many possible integral classes C . Each contains at most one irreducible curve: two distinct curves in that class would have intersection -1 , whereas distinct irreducible curves on a smooth surface have nonnegative intersection. This proves finiteness of the floating curves.

Any presentation of a fixed surface S_P as the blow-up of eight distinct smooth points of Q selects an ordered set of eight disjoint floating curves to contract. There are only finitely many such ordered sets. For each set, the contraction determines the target and the ordered centers up to an isomorphism of the target with Q , that is, up to H . This proves the second assertion.

Finally, $\text{Aut}(S_P)$ permutes the finite set of floating curves. A finite-index subgroup fixes the eight exceptional curves and therefore descends injectively to the stabilizer of the ordered tuple P in H . Indeed, the contraction is an isomorphism away from the eight exceptional curves, so an automorphism inducing the identity after contraction is the identity on their complementary dense open subset, and hence on S_P . If the tuple stabilizer is trivial, this finite-index subgroup is trivial, so $\text{Aut}(S_P)$ is finite. $\square$

We prove that the morphism F assigning hypersurface equations to configurations and graded coordinate changes is dominant by showing that its nonempty fibers have dimension 8. These fibers account for the action of H , without requiring a global moduli space of surfaces.

Proposition A.6 (Dominance of the blow-up construction). *The locus in U of hypersurfaces isomorphic to S_P for some $P \in W$ contains a dense open subset of U . The same conclusion holds after replacing W by any nonempty open subset of it.*

Proof. Choose a nonempty open subset $W_0 \subset W$ on which the tuple stabilizers are trivial, as in Lemma A.4. We first explain how to choose hypersurface equations algebraically on a further nonempty open subset of W_0 . Blowing up the eight tautological sections of $Q \times W_0$ gives an algebraic family $\pi : \mathcal{S} \rightarrow W_0$ with fiber S_P over P . For $m \geq 0$, put

$$\mathcal{L}_m := \mathcal{O}_{\mathcal{S}}(-mK_{\mathcal{S}/W_0})^{\vee\vee}.$$

The eight centers lie in the smooth locus of Q , so near the fixed quotient singularity the family is unchanged from $Q \times W_0$; away from that locus, the blow-up family is smooth over W_0 . Thus the sheaves $\mathcal{L}_m$ are flat over W_0 and restrict to $\mathcal{O}_{S_P}(-mK_{S_P})$ on each fiber. For every $m \geq 0$, Kawamata–Viehweg vanishing for klt surfaces [25, Theorem 2.70], applied to the reflexive sheaf $\mathcal{O}_{S_P}(-mK_{S_P})$, gives

$$H^1(S_P, \mathcal{O}_{S_P}(-mK_{S_P})) = 0,$$

since $-(m+1)K_{S_P}$ is ample. By cohomology and base change [18, Chapter III, Theorem 12.11], after shrinking W_0 the sheaves $\mathcal{E}_m := \pi_* \mathcal{L}_m$, $0 \leq m \leq 10$, are locally free and their formation commutes with base change. Multiplication in the relative anticanonical algebra gives morphisms $\mathcal{E}_a \otimes \mathcal{E}_b \rightarrow \mathcal{E}_{a+b}$, for $a+b \leq 10$, compatible with multiplication in every fiber.

By Lemma 5.6, each fiber has one new anticanonical-ring generator in each of the degrees 1, 2, 3, 5, and no additional generators in the relevant lower degrees. For $m \in \{1, 2, 3, 5\}$, let $\mathcal{D}_m \subset \mathcal{E}_m$ be the image of the multiplication maps generated by lower-degree sections. The fiber dimension of $\mathcal{E}_m/\mathcal{D}_m$ is one; after shrinking W_0 , these quotients are line bundles. Choose local frames and lift them to local sections x, y, z, w of degrees 1, 2, 3, 5, respectively. The chosen generators induce a degree-10 evaluation map

$$\mathcal{O}_{W_0}[x, y, z, w]_{10} \longrightarrow \mathcal{E}_{10}.$$

On every fiber its kernel has dimension one, since Lemma 5.6 gives a single relation of weighted degree 10. After a further shrinking, the constant-rank property makes this kernel a line bundle. Trivializing it locally gives a relation $f_P(x, y, z, w) = 0$ whose coefficients vary regularly with P . Changing the trivialization multiplies all coefficients by the same nowhere-vanishing scalar, so the projective class $[f_P]$ is independent of this choice. Hence these coefficients define a morphism

$$h : W_0 \longrightarrow U, \quad V(h(P)) \simeq S_P,$$

where $V(h(P))$ denotes the hypersurface defined by the projective form $h(P)$.

Allowing all graded coordinate changes gives the morphism

$$F : G \times W_0 \longrightarrow U, \quad (g, P) \longmapsto g \cdot h(P).$$

Its image consists precisely of the equations of hypersurfaces isomorphic to some S_P , $P \in W_0$, by Lemma A.2. We compute the dimension of each nonempty fiber of F .

Fix $u \in F(G \times W_0)$. Projection of $F^{-1}(u)$ to W_0 has image

$$\{P \in W_0 : S_P \simeq V(u)\}.$$

By Lemma A.5, this is a finite union of intersections of H -orbits with W_0 . Since the tuple stabilizer is trivial and $\dim H = 8$, the orbit-dimension formula [8, Proposition I.1.8] gives dimension 8 for each orbit. Every nonempty intersection with the open subset W_0 therefore also has dimension 8. For each tuple in this image, the choices of g satisfying $g \cdot h(P) = u$ form a coset of the stabilizer of $h(P)$. This stabilizer is finite by Lemmas A.3 and A.5. The projection therefore has finite fibers, and $\dim F^{-1}(u) = 8$.

The fiber-dimension theorem now gives

$$\dim F(G \times W_0) = \dim G + \dim W_0 - 8 = \dim U.$$

Thus F is dominant. By Chevalley's theorem [32, Tag 054K], its image is constructible; since it is dense in the irreducible variety U , it contains a dense open subset [32, Tag 005K]. For any nonempty open subset $W' \subset W$, replace W_0 by the nonempty open intersection $W_0 \cap W'$, which is nonempty because W is irreducible; the same dimension and fiber argument proves the final assertion. $\square$

Proof of Proposition 5.7. By Proposition A.6, a general hypersurface in U is isomorphic to S_P with eight distinct general smooth centers. Since $-K_{S_P}$ is ample, S_P is its anticanonical model. The minimal resolution $\tau : \mathbb{F}_3 \rightarrow Q$ is an isomorphism over the smooth locus of Q . The eight centers therefore lift uniquely to eight distinct points on $\mathbb{F}_3$ away from C_0 . Blowing up these points resolves S_P ; its exceptional locus over the singular point is still the single (-3) -curve C_0 , so this resolution is minimal. $\square$

B Brauer groups and obstructions in the Corti–Heuberger cascades

The proof of Theorem 1.2 combines the cascade description of Section 5, the cokernel criterion of Corollary 4.9, and invariance under blow-ups at smooth points (Lemma B.5).

We first describe the lattice mechanism for a normal projective rational surface X with a finite set of singularities p_i of types $\frac{1}{n_i}(1, 1)$, where $n_i \geq 2$. Its minimal resolution has pairwise disjoint exceptional curves $E_1, \dots, E_r$ with $E_i^2 = -n_i$. Put $\Lambda = \bigoplus_i \mathbb{Z}E_i$. In the notation of Lemma 4.8, the intersection map is

$$\theta : \text{Pic}(\tilde{X}) \longrightarrow \Lambda^\vee, \quad D \longmapsto (D \cdot E_1, \dots, D \cdot E_r).$$

Here $\Lambda^\vee = \text{Hom}(\Lambda, \mathbb{Z})$ has basis $e_1, \dots, e_r$ dual to the E_i ; in particular, $\theta(E_i) = -n_i e_i$. By Corollary 4.9, $\text{Br}(X) \cong \text{coker}(\theta)$.

The four-point calculation below instead uses a surface in $X_{4,4/3}$, obtained from $Z_4 = X_{4,7/3}$ by one blow-up at a smooth point.

Bright's exact sequence applies to arbitrary rational surface singularities [9, Proposition 1], whereas the classification carried out in [9, Section 3] concerns the classical singular del Pezzo case obtained by contracting (-2) -curves. In the Corti–Heuberger cascades the exceptional curves over the singular points have self-intersection -3 .

B.1 Saturation of the exceptional lattice

The intersection lattice $P = \text{Pic}(\tilde{X})$ is free and unimodular. Indeed, the rational-surface description used in Corollary 2.6 reduces this to $\mathbb{P}^2$ and the Hirzebruch surfaces, whose intersection matrices have determinant ± 1 . Each point blow-up adds an orthogonal rank-one summand of square -1 . In particular, intersection identifies P with $P^\vee$.

Definition B.1. The *saturation* of the exceptional lattice in P is

$$\Lambda^{\text{sat}} = (\Lambda \otimes \mathbb{Q}) \cap \text{Pic}(\tilde{X}).$$

The lattice is *primitive* if P/Λ is torsion-free, equivalently if $\Lambda^{\text{sat}} = \Lambda$. The finite quotient $\Lambda^{\text{sat}}/\Lambda$ is its *saturation quotient*. Using the intersection form to identify $\Lambda^\vee$ with a lattice in $\Lambda \otimes \mathbb{Q}$, the *discriminant group*

$$A_\Lambda := \Lambda^\vee/\Lambda \cong \bigoplus_{i=1}^r \mathbb{Z}/n_i$$

carries the induced discriminant bilinear form

$$b_\Lambda : A_\Lambda \times A_\Lambda \longrightarrow \mathbb{Q}/\mathbb{Z}, \quad b_\Lambda(\bar{x}, \bar{y}) = x \cdot y \pmod{\mathbb{Z}}.$$

An *integral overlattice* of Λ is a finite-index extension inside $\Lambda \otimes \mathbb{Q}$ whose pairings are integral. It lies in $\Lambda^\vee$, and its quotient by Λ is totally isotropic for b_Λ , meaning that every pairing between elements of the quotient is zero. Conversely, the inverse image of such a subgroup is an integral overlattice. This is the bilinear version of the overlattice correspondence discussed in [27, Section 1.4]. In particular, $H_\Lambda := \Lambda^{\text{sat}}/\Lambda \subset A_\Lambda$ is totally isotropic, since its representatives belong to the integral lattice P .

The next lemma identifies the Brauer group canonically with the dual of the saturation quotient. Nontrivial $\text{Br}(X)$, equivalently torsion in $\text{Cl}(X)$, is the Brauer obstruction to an untwisted adherent KKS decomposition: by Theorem 4.7, such a decomposition requires $\text{Cl}(X)$ to be torsion-free.

Lemma B.2 (The Brauer group and the saturation quotient). *There are natural isomorphisms*

$$\begin{aligned} \text{Br}(X) &\cong \text{coker}(\theta) \cong \text{Hom}(\Lambda^{\text{sat}}/\Lambda, \mathbb{Q}/\mathbb{Z}) \\ &\cong \text{Hom}(\text{Cl}(X)_{\text{tors}}, \mathbb{Q}/\mathbb{Z}). \end{aligned}$$

Thus $\text{Br}(X) = 0$ if and only if Λ is primitive in P . As abstract finite abelian groups, $\text{Br}(X)$ and $\Lambda^{\text{sat}}/\Lambda$ are isomorphic, but this last identification is not canonical.

Proof. Applying $\text{Hom}(-, \mathbb{Z})$ to $0 \rightarrow \Lambda \rightarrow P \rightarrow P/\Lambda \rightarrow 0$ gives an exact sequence

$$P^\vee \longrightarrow \Lambda^\vee \longrightarrow \text{Ext}^1(P/\Lambda, \mathbb{Z}) \longrightarrow 0,$$

because P is free. Under the unimodular identification $P \cong P^\vee$, the first map is θ . For any finitely generated abelian group G , the sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$ gives $\text{Ext}^1(G, \mathbb{Z}) \cong \text{Hom}(G_{\text{tors}}, \mathbb{Q}/\mathbb{Z})$. Pushforward of divisors identifies P/Λ with $\text{Cl}(X)$: every prime divisor on X has a strict transform on the smooth surface $\tilde{X}$, and the kernel is generated by the contracted curves. Hence $(P/\Lambda)_{\text{tors}} = \Lambda^{\text{sat}}/\Lambda \cong \text{Cl}(X)_{\text{tors}}$. The result follows from Corollary 4.9; a finite abelian group is noncanonically isomorphic to its character group. $\square$

The Gram matrix $\text{diag}(-n_1, \dots, -n_r)$ gives $\Lambda^\vee = \bigoplus_i \mathbb{Z}(E_i/n_i)$. Every class in the saturation quotient thus has a representative

$$L = \sum_{i=1}^r \frac{a_i}{n_i} E_i \in \text{Pic}(\tilde{X}), \quad \bar{a}_i \in \mathbb{Z}/n_i,$$

where the a_i are integer lifts of the residue classes. Since the intersection form is integral and adjunction gives $K_{\tilde{X}} \cdot E_i = n_i - 2$, every such representative satisfies

$$L^2 = - \sum_i \frac{a_i^2}{n_i} \in \mathbb{Z}, \quad K_{\tilde{X}} \cdot L = \sum_i \frac{a_i(n_i - 2)}{n_i} \in \mathbb{Z}. \quad (6)$$

These are necessary numerical constraints on classes in the saturation quotient, not a sufficiency criterion for their realization in $\text{Pic}(\tilde{X})$.

For the remainder of this appendix, specialize to k points all of type $\frac{1}{3}(1, 1)$, so $r = k$, all $n_i = 3$, and $A_\Lambda \cong (\mathbb{Z}/3)^k$. Corti–Heuberger’s defect is $\delta_{\text{CH}} = \dim_{\mathbb{F}_3}(\Lambda^{\text{sat}}/\Lambda)$. They determine it in [11, Section 2.1, Proposition 18(b), and Lemma 28]: $\delta_{\text{CH}} = 1$ for

$X_{6,2}, X_{6,1}$ and $\delta_{\text{CH}} = 0$ otherwise. Together with [Lemma B.2](#), they already give the group orders in [Theorem 1.2](#). The calculations below exhibit the intersection maps and classes in the saturation quotient realizing these values. The next lemma specializes (6) and proves primitivity for one or two exceptional curves without choosing a blow-up presentation.

Lemma B.3. *Every class in $\Lambda^{\text{sat}}/\Lambda$ can be represented in the form*

$$L = \frac{1}{3} \sum_{i=1}^k a_i E_i, \quad a_i \in \{0, \pm 1\}.$$

Such a representative satisfies the congruences

$$\sum_{i=1}^k a_i^2 \equiv 0 \pmod{3}, \quad \sum_{i=1}^k a_i \equiv 0 \pmod{3}.$$

Consequently:

- (i) *if $k = 1$ or $k = 2$, the lattice Λ is primitive;*
- (ii) *if $3 \leq k \leq 5$, every nonzero class in the saturation quotient is represented by*

$$\pm \frac{E_a + E_b + E_c}{3}$$

for a triple $\{a, b, c\}$, and the isotropic subgroup $\Lambda^{\text{sat}}/\Lambda$ has order at most 3;

- (iii) *if $k = 6$, classes supported on three or on six exceptional curves are numerically possible. On Z_6 , the saturation quotient is generated by the alternating support-six class, so no support-three class is realized; see [Appendix B.4](#).*

Proof. For $n_i = 3$, choose the residue representatives $a_i \in \{0, \pm 1\}$. Formula (6) becomes

$$-\frac{1}{3} \sum_i a_i^2 \in \mathbb{Z}, \quad \frac{1}{3} \sum_i a_i \in \mathbb{Z},$$

which gives the two stated congruences.

The first congruence says that the number of nonzero coefficients is divisible by 3. This proves (i). If $3 \leq k \leq 5$, a nonzero class has support of size 3, and the second congruence forces its three coefficients to have the same sign. By the cited overlattice correspondence, the isotropic subgroup is totally isotropic for b_Λ . After changing their signs, two support-three classes with distinct supports A and B have pairing

$$-\frac{\#(A \cap B)}{3} \pmod{\mathbb{Z}}.$$

For two distinct triples in a set of at most five elements, the intersection has size 1 or 2, so the two classes cannot be orthogonal. Hence all nonzero classes in the saturation quotient generate the same cyclic subgroup, proving (ii). For $k = 6$, supports of size 3 and 6 satisfy the first numerical condition, subject also to the stated congruence on the signs. In particular, the alternating coefficient vector satisfies both congruences. Thus the numerical conditions permit the coefficient pattern $\frac{1}{3}(E_1 - E_2 + E_3 - E_4 + E_5 - E_6)$, as asserted in (iii). $\square$

Corollary B.4. *Let X be a log del Pezzo surface with at most two singular points, all of type $\frac{1}{3}(1,1)$. Its Brauer group is trivial. In particular, $\mathrm{Br}(Z_1) = \mathrm{Br}(\mathbb{P}(1,1,3)) = 0$ and $\mathrm{Br}(Z_2) = 0$.*

Proof. Combine Lemmas B.2 and B.3 (for $k = 0$, this is $\mathrm{Br}(Y) = 0$, as in Lemma 4.8). $\square$

The bound of two singularities in Corollary B.4 uses the (-3) -lattice congruences. For $k = 1$, this recovers the direct fiber-class computation following Corollary 4.9. For $3 \leq k \leq 5$, the same lemmas leave only the possibilities 0 and $\mathbb{Z}/3$. The intersection maps computed below decide between them.

B.2 Behavior under cascade blow-ups

The morphisms in a Corti–Heuberger cascade are blow-ups at smooth points by [11, Corollary 8]. Their effect on the intersection map is elementary.

Lemma B.5. *Let X be a normal projective surface with quotient singularities and minimal resolution $Y \rightarrow X$, where Y is a rational surface, and let $X' \rightarrow X$ be the blow-up of a smooth point. This blow-up preserves the Brauer group:*

$$\mathrm{Br}(X') \cong \mathrm{Br}(X).$$

The same conclusion holds for a finite sequence of blow-ups at smooth points, including infinitely near centers.

Proof. On minimal resolutions, the induced morphism

$$\beta : \tilde{X}' \longrightarrow \tilde{X}$$

is the blow-up of a point disjoint from the exceptional curves $E_1, \dots, E_k$ of $\tilde{X} \rightarrow X$, since the resolution is an isomorphism over the smooth locus. The strict transforms of these curves give natural identifications

$$\Lambda_{X'} \cong \Lambda_X, \quad \Lambda_{X'}^\vee \cong \Lambda_X^\vee.$$

If D is the new exceptional curve, the Picard group decomposes as

$$\mathrm{Pic}(\tilde{X}') = \beta^* \mathrm{Pic}(\tilde{X}) \oplus \mathbb{Z}D, \quad D \cdot E_i = 0.$$

This decomposition implies that the intersection numbers satisfy

$$(\beta^* L + nD) \cdot E_i = L \cdot E_i$$

for every $L \in \mathrm{Pic}(\tilde{X})$ and every i . Under the displayed identification of the dual exceptional lattices, the image and cokernel of θ are therefore unchanged. The Brauer groups are these cokernels by Corollary 4.9; naturality of Bright’s sequence identifies this isomorphism with pullback. The assertion for a sequence follows step by step, even when a center lies on a curve introduced by an earlier blow-up: that curve lies over the smooth locus, not over a singular point of the current surface. $\square$

Combining this lemma with the marked cascade presentations supplied by [11, Corollary 8] gives, for every $X \in X_{k,d}$,

$$\mathrm{Br}(X) \cong \mathrm{Br}(Z_k). \tag{7}$$

It is therefore enough to compute the cokernel on the rigid base Z_k , or on a surface connected to it by blow-ups at smooth points.

B.3 The cases $k = 3, 4, 5$

Three points: a toric surface. The three-point base admits a direct toric calculation.

Corti–Heuberger exhibit polygon 20 as a toric $\mathbb{Q}$ -Gorenstein degeneration in the family $X_{3,5}$ [11, Section 7, Table 4 and Figure 11]. We check its singularities directly before using uniqueness of the base. Let $N \cong \mathbb{Z}^2$, and let $v_1, \dots, v_5 \in N$ be the primitive ray generators of its fan, in counterclockwise order, with V_i the corresponding torus-invariant divisors:

$$v_1 = (1, 0), \quad v_2 = (-1, 3), \quad v_3 = (-2, 3), \quad v_4 = (-1, 0), \quad v_5 = (0, -1).$$

The first three two-dimensional cones have determinant 3, while the remaining two are smooth. The primitive rays inserted to resolve the three singular cones are

$$u_{12} = (0, 1), \quad u_{23} = (-1, 2), \quad u_{34} = (-1, 1).$$

Let E_1, E_2, E_3 be the corresponding exceptional torus-invariant curves. Each inserted ray is one third of the sum of its original neighbors, and the determinants of all adjacent rays in the subdivided fan equal one. Thus the subdivision is the minimal toric resolution, obtained by the ray-subdivision procedure of [13, Chapters 10–11]. Computing the boundary self-intersections from the fan relations as in [13, Theorem 10.4.4] gives the following table; the last column records $w_{i-1} + w_{i+1} = b_i w_i$, with square $-b_i$ for the corresponding curve:

curve	square	ray relation
V_1	0	$v_5 + u_{12} = 0$
E_1	-3	$v_1 + v_2 = 3u_{12}$
V_2	-1	$u_{12} + u_{23} = v_2$
E_2	-3	$v_2 + v_3 = 3u_{23}$
V_3	-1	$u_{23} + u_{34} = v_3$
E_3	-3	$v_3 + v_4 = 3u_{34}$
V_4	-1	$u_{34} + v_5 = v_4$
V_5	0	$v_4 + v_1 = 0$

The inserted curves are therefore precisely the exceptional (-3) -curves, so the surface has exactly three $\frac{1}{3}(1, 1)$ -points and no other singularities. After the unimodular change of coordinates $(x, y) \mapsto (x, x + y)$, these are the rays of the polygon cited above. It belongs to $X_{3,5}$ (no higher Fano index family has three singular points) and hence is isomorphic to Z_3 by the uniqueness statement recalled in Section 5.

The boundary intersections give

$$\theta(V_1) = e_1, \quad \theta(V_2) = e_1 + e_2, \quad \theta(V_4) = e_3.$$

Consequently, the dual basis vectors satisfy

$$e_1 = \theta(V_1), \quad e_2 = \theta(V_2) - \theta(V_1), \quad e_3 = \theta(V_4),$$

so θ is surjective and the Brauer group vanishes:

$$\mathrm{Br}(Z_3) = \mathrm{Br}(X_{3,5}) = 0.$$

Equivalently, the divisor classes

$$\mathcal{O}(-V_1), \quad \mathcal{O}(V_1 - V_2), \quad \mathcal{O}(-V_4)$$

have intersection $-\delta_{ij}$ with E_j .

Corollary B.6 (KKS decomposition in the three-point cascade). *Let $X \in X_{3,d}$ and $s = 5 - d$. Then*

$$D^b(\text{coh } X) = \left\langle D^b(K(3,1)\text{-mod}), D^b(K(3,1)\text{-mod}), D^b(K(3,1)\text{-mod}), F_1, \dots, F_{2+s} \right\rangle,$$

where every F_i is exceptional.

Proof. The toric base Z_3 has torsion-free class group. Of its five fixed points, the first three in the displayed fan are quotient singularities and the last two are smooth. KKS [23, Proposition 5.6, Remark 5.8, and Corollary 5.10] therefore gives three untwisted adherent subcategories on the resolution and two components generated by single exceptional objects. Here M in KKS Remark 5.8 is the correcting line bundle on $\tilde{Z}_3$ used in tensoring with $M \otimes \mathcal{O}(K_{\tilde{Z}_3})$: its degree is zero on every exceptional curve except the final curve over the last fixed point, where it is -1 . Since that fixed point is smooth, the exception is absent and one takes $M = \mathcal{O}$. By [11, Corollary 8(3)], X is obtained by s blow-ups at smooth points of Z_3 , at distinct centers by (3). On resolutions these avoid all three (-3) -curves. Lemma 4.4 preserves the three adherent subcategories under pullback, and Orlov's decomposition with $\mathcal{O}_D$ on the right adds s exceptional objects. KKS descent and the identification of the descended adherent components with $D^b(K(3,1)\text{-mod})$ [23, Theorems 2.12(i) and 3.16] give the stated decomposition. $\square$

Remark B.7 (Toric surfaces and toric degenerations). Among nonsmooth del Pezzo surfaces with only $\frac{1}{3}(1,1)$ -points, the toric surfaces are precisely those in

$$S_{1,25/3}, \quad X_{1,22/3}, \quad X_{1,19/3}, \quad X_{3,5}, \quad X_{3,4}, \quad X_{6,2}.$$

These are the toric cases in the Corti–Heuberger classification [11, Tables 1, 2, and 4]; see also [28, Table 1]. In particular, Z_k is toric precisely for $k \in \{1, 3, 6\}$.

The toric degenerations of Z_2, Z_4, Z_5 in [11, Table 4, polygons 23, 11, and 7] have additional T -singularities: respectively an A_1 -point, an A_2 -point, and two $\frac{1}{4}(1,1)$ -points. Thus these toric special fibers are not the rigid bases themselves. The semiorthogonal decompositions for toric surfaces in [23, Section 5] describe the derived categories of these special fibers.

The k -gon construction. For the four- and five-point calculations we use the k -gon construction. It starts with a smooth del Pezzo surface of degree k containing a cyclic configuration $C_1 + \dots + C_k$ of (-1) -curves, blows up the vertices $C_i \cap C_{i+1}$, and contracts the strict transforms of the C_i . Corti–Heuberger show that the resulting surface has k singularities of type $\frac{1}{3}(1,1)$, degree $k/3$, and belongs to the family $X_{k,k/3}$ [11, Section 3.5, “ k -gons”]. Thus the surfaces used below are respectively members of $X_{4,4/3}$ and $X_{5,5/3}$. Let D_i be the exceptional curve over the i -th vertex and E_i the strict transform of C_i , with cyclic indices. Direct intersection calculations give $E_i^2 = -3$, since each C_i passes through two vertices. The curves E_i are pairwise disjoint, and each D_i meets just E_i and E_{i+1} , transversally once. Consequently,

$$\theta(D_i) = e_i + e_{i+1}, \quad \theta(E_i) = -3e_i.$$

The degree also follows from the discrepancy computation in the proof of Corollary 3.6: the k blow-ups lower K^2 from k to zero, and contraction of the k disjoint (-3) -curves raises it to $k/3$.

Five points. For the five-point base, consider the degree-five del Pezzo surface $\text{Bl}_{p_1, \dots, p_4} \mathbb{P}^2$, with exceptional classes Q_i and line class H . The curves

$$Q_1, \quad H - Q_1 - Q_2, \quad Q_2, \quad H - Q_2 - Q_3, \quad H - Q_1 - Q_4$$

have adjacent intersections one and nonadjacent intersections zero, so they form a cyclic pentagon of (-1) -curves. Blowing up its vertices and contracting the strict transforms of its sides gives a member of $X_{5,5/3}$, which is the rigid base Z_5 by uniqueness. The five vectors $e_i + e_{i+1}$ are the columns of

$$A_5 = I + P_5,$$

where P_5 is the cyclic permutation matrix. A direct calculation gives

$$\det(A_5) = \prod_{\zeta^5=1} (1 + \zeta) = 2.$$

The quotient $\mathbb{Z}^5/A_5\mathbb{Z}^5$ has order 2. Since the image of θ also contains $3\mathbb{Z}^5$, and multiplication by 3 is invertible on this order-two quotient, we have

$$A_5\mathbb{Z}^5 + 3\mathbb{Z}^5 = \mathbb{Z}^5.$$

Hence θ is surjective and the Brauer group vanishes:

$$\text{Br}(Z_5) = \text{Br}(X_{5,5/3}) = 0.$$

Four points. For the four-point cascade, it is convenient to use a member of $X_{4,4/3}$, obtained from the rigid base $Z_4 = X_{4,7/3}$ by one blow-up at a smooth point [11, Corollary 8]. By Lemma B.5, it has the same cokernel as Z_4 . The vectors $e_i + e_{i+1}$ span a rank-three subgroup, so one additional divisor class is needed. To avoid confusing the exceptional classes on a degree-four del Pezzo surface with the (-3) -curves above, write

$$S = \text{Bl}_{p_1, \dots, p_5} \mathbb{P}^2$$

with basis $H, Q_1, \dots, Q_5$, where $H^2 = 1$, $Q_i^2 = -1$, and all other pairings between basis elements vanish. Consider

$$\begin{aligned} C_1 &= Q_1, \\ C_2 &= H - Q_1 - Q_2, \\ C_3 &= Q_2, \\ C_4 &= 2H - Q_1 - Q_2 - Q_3 - Q_4 - Q_5. \end{aligned}$$

Their self-intersections are all -1 . The adjacent intersections are

$$C_1 \cdot C_2 = C_2 \cdot C_3 = C_3 \cdot C_4 = C_4 \cdot C_1 = 1,$$

and the nonadjacent intersections are

$$C_1 \cdot C_3 = 0, \quad C_2 \cdot C_4 = 0.$$

Their intersection graph is therefore a 4-cycle. In addition,

$$Q_3 \cdot C_1 = Q_3 \cdot C_2 = Q_3 \cdot C_3 = 0, \quad Q_3 \cdot C_4 = 1.$$

For general $p_1, \dots, p_5$, these intersections are transverse; in particular, Q_3 avoids the four vertices of the cycle. If π is their blow-up, the projection formula gives

$$\theta(\pi^*Q_3) = e_4.$$

Using this as the starting dual basis vector, the remaining basis vectors are obtained successively:

$$e_1 = \theta(D_4) - e_4,$$

$$e_2 = \theta(D_1) - e_1,$$

$$e_3 = \theta(D_2) - e_2.$$

These identities make θ surjective. Since the 4-gon representative is a one-point blow-up of Z_4 , [Lemma B.5](#) gives

$$\mathrm{Br}(Z_4) = \mathrm{Br}(X_{4,7/3}) \cong \mathrm{Br}(X_{4,4/3}) = 0.$$

B.4 The six-point Brauer obstruction

Unlike the five smaller cascades, the six-point cascade carries a nonzero Brauer obstruction. We first exhibit a relation among the toric boundary divisors that produces an order-three class in the saturation quotient $\Lambda^{\mathrm{sat}}/\Lambda \subset A_\Lambda$. This proves that $\mathrm{Br}(Z_6) \neq 0$. We then compute the full image of the intersection map: a single alternating-sum congruence modulo three describes it, showing that the Brauer group is exactly $\mathbb{Z}/3$.

For the rigid base $Z_6 = X_{6,2}$ of [Section 5](#), Corti–Heuberger describe its fan in [\[11, Section 2.2.5\]](#). We compute the intersection map by first describing the minimal toric resolution and then determining the lattice generated by the images of its boundary divisors. Let $N \cong \mathbb{Z}^2$, and let $v_1, \dots, v_6 \in N$ be the primitive ray generators of its fan, with V_i the corresponding torus-invariant divisors:

$$\begin{aligned} v_1 &= (1, 0), & v_2 &= (-1, 3), & v_3 &= (-2, 3), \\ v_4 &= (-1, 0), & v_5 &= (1, -3), & v_6 &= (2, -3), \end{aligned}$$

in counterclockwise order. The unimodular change $(x, y) \mapsto (x, x+y)$ sends these vertices to those of [\[11, Figure 1\]](#). Indices are cyclic modulo 6 throughout. Every adjacent cone has determinant 3, and hence defines a cyclic quotient singularity of order 3. The primitive rays inserted in the minimal toric resolution are

$$u_i = \frac{v_i + v_{i+1}}{3}, \quad i \in \mathbb{Z}/6\mathbb{Z}.$$

The inserted rays are explicitly

$$\begin{aligned} u_1 &= (0, 1), & u_2 &= (-1, 2), & u_3 &= (-1, 1), \\ u_4 &= (0, -1), & u_5 &= (1, -2), & u_6 &= (1, -1). \end{aligned}$$

Let E_i be the exceptional torus-invariant curve corresponding to u_i . Since

$$\det(v_i, u_i) = \det(u_i, v_{i+1}) = 1,$$

the subdivision is smooth and gives the minimal toric resolution [\[13, Chapters 10–11\]](#). The resolved boundary is the alternating cycle

$$V_1, E_1, V_2, E_2, \dots, V_6, E_6,$$

and the ray relations are

$$u_{i-1} + u_i = v_i, \quad v_i + v_{i+1} = 3u_i.$$

Therefore, [13, Theorem 10.4.4] gives

$$V_i^2 = -1, \quad E_i^2 = -3.$$

Consequently, the minimal resolution of each singularity consists of one (-3) -curve, so all six singularities are of type $\frac{1}{3}(1, 1)$.

A nonzero class in the saturation quotient. An order-three class in the saturation quotient comes from a divisibility relation. The principal-divisor formula $\text{div}(\chi^m) = \sum_\rho \langle m, v_\rho \rangle D_\rho$ of [13, Theorem 4.1.3], applied to the character $m = (0, 1)$, gives

$$\begin{aligned} 0 &= E_1 + 3V_2 + 2E_2 + 3V_3 + E_3 - E_4 \\ &\quad - 3V_5 - 2E_5 - 3V_6 - E_6. \end{aligned}$$

Rearranging gives

$$E_1 - E_2 + E_3 - E_4 + E_5 - E_6 = 3L, \tag{8}$$

where the divisor L is

$$L = -V_2 - V_3 + V_5 + V_6 - E_2 + E_5 \in \text{Pic}(\tilde{Z}_6).$$

Thus L lies in Λ^{sat} but not in Λ : the E_i are linearly independent, and its coefficients in this basis are $\pm 1/3$. Its class has order three and realizes the alternating pattern allowed by Lemma B.3. Moreover, $L^2 = -2$, confirming the isotropy of its class for the discriminant pairing in Definition B.1. In particular, Lemma B.2 already implies that $\text{Br}(Z_6) \neq 0$. We next determine the whole group.

The cokernel. For a complete toric variety, the torus-invariant prime divisors generate $\text{Cl}(\tilde{Z}_6)$, and smoothness gives $\text{Cl}(\tilde{Z}_6) = \text{Pic}(\tilde{Z}_6)$ [13, Theorem 4.1.3]. Hence the toric boundary generates $\text{Pic}(\tilde{Z}_6)$, and its image under θ is generated by

$$\theta(E_i) = -3e_i, \quad \theta(V_j) = e_{j-1} + e_j,$$

with cyclic indices. Put $c_j = e_{j-1} + e_j$, where $e_0 = e_6$, and let $A \subset \mathbb{Z}^6$ be the subgroup generated by $c_2, \dots, c_6$. These five vectors, together with e_6 , form a basis of $\mathbb{Z}^6$: the matrix of their columns is lower triangular with diagonal entries one. Thus A is primitive of rank five. Moreover,

$$c_1 = c_2 - c_3 + c_4 - c_5 + c_6,$$

so A is also the span of all six images $\theta(V_j)$. Consider the surjective homomorphism

$$\lambda : \mathbb{Z}^6 \longrightarrow \mathbb{Z}, \quad (a_1, \dots, a_6) \longmapsto a_1 - a_2 + a_3 - a_4 + a_5 - a_6.$$

Every c_j lies in $\ker(\lambda)$. Since both A and $\ker(\lambda)$ have rank five and A is primitive, this inclusion is an equality. The image of the full toric boundary is therefore

$$\text{im}(\theta) = A + 3\mathbb{Z}^6 = \ker(\lambda) + 3\mathbb{Z}^6.$$

This sum is the kernel of the reduction of λ modulo 3. Indeed, if $\lambda(a) = 3q$, then $a - 3qe_1 \in \ker(\lambda)$, while the reverse inclusion is immediate. Consequently,

$$\mathrm{im}(\theta) = \left\{ (a_1, \dots, a_6) \in \mathbb{Z}^6 : a_1 - a_2 + a_3 - a_4 + a_5 - a_6 \equiv 0 \pmod{3} \right\}.$$

The homomorphism $\lambda \bmod 3 : \mathbb{Z}^6 \rightarrow \mathbb{Z}/3$ is surjective, so the cokernel and Brauer group are

$$\mathrm{coker}(\theta) \cong \mathbb{Z}/3, \quad \mathrm{Br}(Z_6) = \mathrm{Br}(X_{6,2}) \cong \mathbb{Z}/3.$$

By Lemma B.2, the saturation quotient also has order three. It is therefore generated by the class of L in (8); no support-three class in the saturation quotient occurs on Z_6 . Indeed, its two nonzero elements are $[L]$ and $-[L]$, whose coefficient vectors modulo three are $\pm(1, -1, 1, -1, 1, -1)$; both have all six entries nonzero. Here the global toric divisor relations select the alternating support-six class, whereas the computations for $k = 3, 4, 5$ exclude even the numerically permitted support-three classes.

For the adherence convention we use classes with $L_i \cdot E_j = -\delta_{ij}$; this differs from the dual-basis convention of Corollary 4.9 only by replacing each L_i by $-L_i$. We keep this sign convention in the twisted six-point construction below.

The obstruction to untwisted adherence. In particular, there cannot exist divisor classes L_i satisfying

$$L_i \cdot E_j = -\delta_{ij},$$

since such classes would make θ surjective.

By Theorem 4.7, an untwisted adherent decomposition would force $\mathrm{Cl}(X_{6,2})$ to be torsion-free, equivalently $\mathrm{Br}(X_{6,2}) = 0$. The computed nonzero class rules this out. The KKS result for toric surfaces instead gives a semiorthogonal decomposition of a Brauer-twisted category on Z_6 . Under the cascade blow-ups, the Brauer class is pulled back and the corresponding twisted adherent decomposition is preserved, as proved below.

Corollary B.8 (KKS decomposition of the twisted category in the six-point cascade). *Let $X \in X_{6,d}$, let $\eta : X \rightarrow Z_6$ be a marked cascade presentation, and put $s = 2 - d$. There is a generator $\alpha \in \mathrm{Br}(Z_6)$ such that, with $\alpha_X = \eta^* \alpha$,*

$$D^b(\mathrm{coh}(X, \alpha_X)) = \left\langle \underbrace{D^b(K(3, 1)\text{-mod}), \dots, D^b(K(3, 1)\text{-mod})}_{6 \text{ components}}, F_1, \dots, F_s \right\rangle.$$

Here $\mathrm{coh}(X, \alpha_X)$ uses the KKS convention of coherent right modules over an Azumaya algebra representing α_X , and the F_i are exceptional objects of its derived category; their list is empty when $s = 0$. The class α_X is nonzero.

Proof. The proof has three steps: we first identify the twisted adherence data on the resolution of Z_6 , then pull these data through the cascade blow-ups using Lemma 4.4, and finally identify the descended twisted components via the KKS twisted pushforward.

Write $\pi : \tilde{Z}_6 \rightarrow Z_6$ for the resolution and use the counterclockwise boundary order $V_1, E_1, \dots, V_6, E_6$ above. In KKS notation, $E_{i,0} = V_i$, $E_{i,1} = E_i$, $m_i = 1$, and $d_{i,1} = -E_i^2 = 3$; the last fixed point corresponds to the cone between v_6 and v_1 . [23, Proposition 5.6, equations (5.4)–(5.6), p. 510] gives the component $\langle \mathcal{L}_{i,0}, \mathcal{L}_{i,0}(E_i) \rangle$, where

$$\mathcal{L}_{i,0} = \mathcal{O}\left(\sum_{j < i} (V_j + E_j) + V_i\right), \quad \deg(\mathcal{L}_{i,0}|_{E_i}) = 1 + \delta_{i6}.$$

Indeed, V_i meets E_i once, and the sum contributes only $V_1 \cdot E_6 = 1$ when $i = 6$. Our adherence convention $\deg(\mathcal{L}_{i,0}|_{E_i}) = 3 + b_i - 1$ therefore gives $b_i = -1 + \delta_{i6}$, exactly [23, Proposition 5.6, equation (5.7)]. Adjunction gives $K_{\tilde{Z}_6} \cdot E_i = 1$, so tensoring with $\mathcal{O}(K_{\tilde{Z}_6})$ adds one to every adherence parameter:

$$b = (-1, -1, -1, -1, -1, 0), \quad b + \theta(K_{\tilde{Z}_6}) = e_6.$$

We also fix the Brauer sign: KKS's map B sends the degree vector of a splitting bundle $\mathcal{V}$, with $\pi^*\mathcal{R} \cong \text{End}(\mathcal{V})$, to $[\mathcal{R}]$, not $[\mathcal{R}^{\text{op}}]$. The twisted pushforward is $F \mapsto R\pi_*(F \otimes \mathcal{V}^\vee)$, a right $\mathcal{R}$ -module, as in [23, Proposition 4.9 and equations (4.13)–(4.14), pp. 498–500]. Since $B \circ \theta = 0$, the displayed equality gives $\alpha = B(b) = B(e_6)$, agreeing with [23, equation (5.8) and Theorem 5.9]. Finally $\lambda(e_6) = -1$, so this class generates $\text{Br}(Z_6)$.

There are no smooth fixed points, so the decomposition of $D^b(\text{coh}(Z_6, \alpha))$ has six $K(3, 1)$ -components and no components generated by single exceptional objects. Under the blow-ups $\tilde{X} \rightarrow \tilde{Z}_6$, Lemma 4.4 preserves the adherent components and their adherence parameters. Orlov's decomposition with $\mathcal{O}_D$ on the right adds s components generated by single exceptional objects. The intersection-map description in Lemma B.5 identifies the resulting Brauer class with $\eta^*\alpha$, still a generator. More explicitly, a vector bundle on $\tilde{Z}_6$ realizing the adherence parameters, as in [23, Corollary 4.10(1)], pulls back to one with the same restrictions on the six exceptional curves.

Applying the twisted pushforward of [23, equation (4.14)] and then [23, Theorem 4.19] to each adherent component identifies its image with $D^b(K(3, 1)\text{-mod})$. For the additional Orlov objects, the restrictions to the six exceptional curves are zero, so [23, Lemma 4.14] identifies each with the twisted pullback of its pushforward. Full faithfulness of twisted pullback preserves their exceptionality; adjunction preserves their semiorthogonality to the preceding components. Fullness follows from [23, Lemma 4.13] and the full resolution decomposition. Thus the components generated by single exceptional objects require no line-bundle adherence hypothesis. This is twisted descent, not ordinary pushforward under Theorem 4.2. $\square$

B.5 Proof of Theorem 1.2

We first collect the computations on the bases. For $k = 1, 2$, Corollary B.4 gives vanishing without choosing a blow-up presentation. For $k = 3, 5$, Appendix B.3 computes the Brauer group on a toric surface and a pentagon surface in the respective base families. Each is the corresponding Z_k , by uniqueness of the base in Section 5. For $k = 4$, the same subsection uses a quadrilateral surface in $X_{4,4/3}$, not in the base family; its contraction to a smooth point of Z_4 and Lemma B.5 give $\text{Br}(Z_4) = 0$. Finally, the toric computation in Appendix B.4 gives $\text{Br}(Z_6) \cong \mathbb{Z}/3$. Altogether,

$$\text{Br}(Z_k) \cong \begin{cases} 0, & 1 \leq k \leq 5, \\ \mathbb{Z}/3, & k = 6. \end{cases}$$

For every surface X in a Fano index one family $X_{k,d}$, (7) now gives $\text{Br}(X) \cong \text{Br}(Z_k)$.

The auxiliary families $B_{1,16/3}$ and $B_{2,8/3}$ are covered directly by Corollary B.4, since their members have one and two singular points, respectively. Their Brauer groups can also be computed by invariance under blow-ups at smooth points: the alternative sequences of blow-ups and floating contractions in [11, Remark 9], recalled in Section 5, give a blow-up at a smooth point $X' \rightarrow B$ and four floating contractions from X' to Z_k ,

for $k = 1, 2$ respectively. Reversing those contractions and applying [Lemma B.5](#) gives

$$\mathrm{Br}(B) \cong \mathrm{Br}(X') \cong \mathrm{Br}(Z_k) = 0.$$

Together with the already treated base Z_1 , these account for all higher Fano index families.

We have now covered every family in the six cascades. The final assertion follows from [\[23, Corollary 4.8\]](#): the nonzero Brauer group in the six-point cascade obstructs an untwisted adherent decomposition, whereas this Brauer obstruction is absent in the other five cascades. This proves [Theorem 1.2](#). $\square$

C Explicit exceptional collections on the rigid cascade bases

This appendix constructs marked blow-up presentations of the minimal resolutions $\tilde{Z}_k$ and uses Orlov’s formula to write the corresponding exceptional collections, proving [Proposition 5.4](#). The descriptions of the singular surfaces come from Corti–Heuberger and [Appendix B](#); the blow-down sequences and the elementary transformation below express their resolutions as successive point blow-ups. Unlike the exceptional-lattice computations in [Appendix B](#), these presentations specify the ordered centers needed for Orlov’s formula.

After fixing uniform notation, we treat the six bases in order. [Appendices C.1](#) and [C.2](#) begin with $\mathbb{F}_3$ and distinguish the incidence conditions that create a second quotient point. The three- and six-point cases are checked by toric contractions; the four-point case requires an elementary transformation, and the five-point case uses the pentagon construction. Each case ends with its ordered exceptional collection on the resolution, and [Corollary C.3](#) extends these collections along the successive blow-ups of a chosen marked cascade presentation of a surface of Fano index one.

The following guide records the centers and geometric steps for the resolutions in [Table 1](#). All modifications are ordinary blow-ups at smooth points; “infinitely near” describes the location of a center relative to the starting surface Y_0 listed there.

k	Centers	Key geometric step
1	None	Use the negative section.
2	One point and two infinitely near points	Create a second (-3) -curve.
3	One torus-fixed point and three infinitely near torus-fixed points	Reverse four boundary contractions.
4	Five distinct points and two infinitely near points	Perform an elementary transformation.
5	Four plane points, then five vertices (four infinitely near)	Separate the pentagon curves.
6	Four torus-fixed points and four infinitely near torus-fixed points	Contract to a hexagon, then a square.

The individual cases below specify the centers and their order.

We use the following notation throughout. For a sequence of ordinary point blow-ups

$$Y_m \xrightarrow{\beta_m} Y_{m-1} \longrightarrow \cdots \xrightarrow{\beta_1} Y_0, \quad \sigma = \beta_1 \circ \cdots \circ \beta_m,$$

let $D_i \subset Y_i$ be the exceptional curve created by β_i , and set

$$G_i = L\beta_m^* \cdots L\beta_{i+1}^* \mathcal{O}_{D_i}(-1). \tag{9}$$

Thus $G_m = \mathcal{O}_{D_m}(-1)$. If $\langle L_1, \dots, L_t \rangle$ is a full exceptional collection of line bundles on Y_0 , [Theorem 2.5](#) gives the ordered full exceptional collection

$$\langle G_m, \dots, G_1, \sigma^* L_1, \dots, \sigma^* L_t \rangle. \quad (10)$$

In particular, a blow-up center may be infinitely near to an earlier point; the objects in (9) are not replaced by sheaves on the final strict transforms. The collections on Y_0 , recalled in [Section 2](#), are

$$\begin{aligned} \mathcal{B}_{\mathbb{P}^2} &= \langle \mathcal{O}, \mathcal{O}(H), \mathcal{O}(2H) \rangle, \\ \mathcal{B}_{\mathbb{F}_n} &= \langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + nf), \mathcal{O}(C_0 + (n+1)f) \rangle. \end{aligned}$$

Here H is a line class, and C_0, f are the section and fiber classes with $C_0^2 = -n$. We denote the collection on $\tilde{Z}_k$ by $\mathcal{A}_k$, the canonical stack of Z_k by $\mathcal{Z}_k$, and its Ishii–Ueda embedding by Φ_k . In each case, [Corollary 3.6](#) turns $\mathcal{A}_k$ into a full exceptional collection on $\mathcal{Z}_k$ by applying Φ_k and prepending the k local objects. The lengths are recorded in [Table 1](#), so we list only the collections on the resolutions below.

C.1 The one-point base

The minimal resolution of $Z_1 = \mathbb{P}(1, 1, 3)$ is $\mathbb{F}_3$, with exceptional curve C_0 . No point blow-ups are needed, and we take

$$\mathcal{A}_1 = \langle \mathcal{O}, \mathcal{O}(f), \mathcal{O}(C_0 + 3f), \mathcal{O}(C_0 + 4f) \rangle.$$

C.2 The two-point base

The directed minimal model program of [\[11, Section 6\]](#) contains a two-point branch ending at $\mathbb{P}(1, 1, 3)$:

$$Z_2 = X_{2,17/3} \longrightarrow T \longrightarrow \mathbb{P}(1, 1, 3), \quad \text{Sing}(T) = \left\{ \frac{1}{3}(1, 1), A_1 \right\}.$$

For the exceptional collection we need a marked blow-up presentation of the minimal resolution, which we now construct directly.

Proposition C.1. *Choose $q_1 \in \mathbb{F}_3 \setminus C_0$, and let f_0 be its ruling fiber. Blow up q_1 , with exceptional curve D_1 , and then choose $q_2 \in D_1$ away from the strict transform of f_0 . Blow up q_2 , with exceptional curve D_2 . Finally, choose q_3 on the strict transform of D_1 , away from both D_2 and the strict transform of f_0 , and blow it up. Then*

$$\tilde{Z}_2 \simeq Y_3 = \text{Bl}_{q_3} \text{Bl}_{q_2} \text{Bl}_{q_1} \mathbb{F}_3.$$

The two curves contracted to the quotient singularities are the untouched C_0 and the final strict transform A of D_1 .

Proof. Write Y_i for the surface after the first i blow-ups and T_i for the total transform of the class of D_i on Y_3 . Then $T_i T_j = -\delta_{ij}$, and pullbacks from $\mathbb{F}_3$ are orthogonal to all T_i . On Y_2 the strict transform A_2 of D_1 has square -2 , since the second blow-up is centered at $q_2 \in D_1$. The last center lies on A_2 , but on neither of its two neighboring curves. Consequently

$$A = T_1 - T_2 - T_3, \quad A^2 = -3, \quad C_0 A = 0.$$

The strict transform $F' = f - T_1$ of f_0 has square -1 ; it meets C_0 and A once each. The curves D_2, D_3 remain (-1) -curves, each meeting A once and disjoint from C_0 and F' . The dual graph records these transverse intersections (each edge has intersection multiplicity one):

$$\begin{array}{ccccc} & & & D_2(-1) & \\ & & & | & \\ C_0(-3) & \text{---} & F'(-1) & \text{---} & A(-3) \\ & & & | & \\ & & & D_3(-1) & \end{array}$$

To check that contracting C_0 and A gives the required del Pezzo surface, consider the rational divisor

$$P = -K_{Y_3} - \frac{1}{3}(C_0 + A) = \frac{5}{3}C_0 + 5f - \frac{4}{3}T_1 - \frac{2}{3}T_2 - \frac{2}{3}T_3.$$

The intersection form in the orthogonal total-transform basis gives

$$P^2 = \frac{17}{3}, \quad PC_0 = PA = 0, \quad PF' = \frac{1}{3}, \quad PD_2 = PD_3 = \frac{2}{3}.$$

Every other irreducible curve is the strict transform of a fiber not through q_1 , or of a curve $B \sim aC_0 + bf$ with $a \geq 1$: an irreducible curve on $\mathbb{F}_3$ with $a = 0$ is a fiber, and curves exceptional over $\mathbb{F}_3$ were already listed. Since $B \neq C_0$, nonnegative intersection of distinct irreducible curves gives $BC_0 = b - 3a \geq 0$, hence $b \geq 3a$. Let m_1, m_2, m_3 be its multiplicities at the successive centers. Intersection with f_0 gives $m_1 \leq a$, while nonnegative intersection of its final strict transform with $A = T_1 - T_2 - T_3$ gives $\tilde{B}A = m_1 - m_2 - m_3 \geq 0$, hence $m_2 + m_3 \leq m_1$. Consequently,

$$P \cdot \tilde{B} = \frac{5}{3}b - \frac{4}{3}m_1 - \frac{2}{3}(m_2 + m_3) \geq \frac{5}{3}b - 2a \geq 3a > 0.$$

On a fiber not through q_1 , the intersection is $5/3$. Thus P is nef and big, and its only null curves are C_0, A . The pair $(Y_3, (C_0 + A)/3)$ is klt, so the basepoint-free theorem [25, Section 3.2] makes a sufficiently divisible multiple of P semiample. Its associated contraction has precisely these two disjoint (-3) -curves as exceptional locus and gives a normal surface Z with two $\frac{1}{3}(1, 1)$ -points. The discrepancy formula identifies P with the pullback of the ample divisor $-K_Z$.

There is also no ambiguity about the Fano index. The divisor-class group of Z is the quotient of $\text{Pic}(Y_3)$ by C_0, A , so it is free with basis f, T_2, T_3 . In this quotient

$$-K_Z = 5f - 2T_2 - 2T_3$$

is primitive. The surface therefore has Fano index one, and [11, Theorem 4 and Remark 7] identifies it with Z_2 . Since its exceptional curves have square -3 , Y_3 is its minimal resolution. $\square$

With $\sigma_2 : Y_3 \rightarrow \mathbb{F}_3$ the composite, the seven-object collection is

$$\mathcal{A}_2 = \langle G_3, G_2, G_1, \sigma_2^* \mathcal{O}, \sigma_2^* \mathcal{O}(f), \sigma_2^* \mathcal{O}(C_0 + 3f), \sigma_2^* \mathcal{O}(C_0 + 4f) \rangle.$$

C.3 The three-point base

For the three-point base, use the fan and boundary notation of [Appendix B.3](#). Its minimal resolution has boundary $V_1, E_1, V_2, E_2, V_3, E_3, V_4, V_5$, with self-intersections

$$(0, -3, -1, -3, -1, -3, -1, 0).$$

To obtain a marked blow-up presentation over $\mathbb{F}_3$, we contract, in this order, the successive images of V_4, V_3, E_3, E_2 . The following table lists the neighboring rays at the actual stage of each contraction:

curve	neighboring rays	ray relation
V_4	u_{34}, v_5	$v_4 = u_{34} + v_5$
V_3	u_{23}, u_{34}	$v_3 = u_{23} + u_{34}$
E_3	u_{23}, v_5	$u_{34} = u_{23} + v_5$
E_2	v_2, v_5	$u_{23} = v_2 + v_5$

In every row the determinant of the two neighbors is one. The ray-sum criterion therefore verifies that the curve being removed is a (-1) -curve and that its contraction is a smooth toric blow-down [13, Theorem 10.4.4]. The remaining fan has rays v_1, u_{12}, v_2, v_5 , with boundary squares $(0, -3, 0, 3)$; it is the fan of $\mathbb{F}_3$, with negative section corresponding to u_{12} .

Reversing these four contractions prescribes the centers: insert u_{23} in the cone $\langle v_2, v_5 \rangle$, then u_{34} in $\langle u_{23}, v_5 \rangle$, then v_3 in $\langle u_{23}, u_{34} \rangle$, and finally v_4 in $\langle u_{34}, v_5 \rangle$. Each center is the intersection of the two indicated boundary curves. This gives $\sigma_3 : \tilde{Z}_3 \rightarrow \mathbb{F}_3$ and the full exceptional collection

$$\mathcal{A}_3 = \langle G_4, G_3, G_2, G_1, \sigma_3^* \mathcal{O}, \sigma_3^* \mathcal{O}(f), \sigma_3^* \mathcal{O}(C_0 + 3f), \sigma_3^* \mathcal{O}(C_0 + 4f) \rangle.$$

The creation stages for G_1, G_2, G_3, G_4 correspond respectively to the rays u_{23}, u_{34}, v_3, v_4 . In particular, G_1 and G_2 are understood as derived pullbacks from their creation stages, not as sheaves on the final (-3) -curves.

C.4 The four-point base

The surface obtained from the quadrilateral in Appendix B.3 lies in $X_{4,4/3}$, not in the rigid base family $Z_4 = X_{4,7/3}$. The following proposition contracts an additional (-1) -curve and performs an elementary transformation of $\mathbb{F}_1$ to give a marked blow-up presentation of $\tilde{Z}_4$ over $\mathbb{F}_2$.

Proposition C.2. *Let $S = \text{Bl}_{p_1, \dots, p_5} \mathbb{P}^2$, with the points general, and let $H, Q_1, \dots, Q_5$ be its standard divisor classes. Set*

$$\begin{aligned} C_1 &= Q_1, & C_2 &= H - Q_1 - Q_2, \\ C_3 &= Q_2, & C_4 &= 2H - Q_1 - Q_2 - Q_3 - Q_4 - Q_5. \end{aligned}$$

Blow up the vertices $r_i = C_i \cap C_{i+1}$, cyclically, with exceptional curves R_i , obtaining Y . The strict transform of the line tangent at p_1 to the conic through the five points is a (-1) -curve

$$D = H - Q_1 - R_4.$$

Its contraction $Y \rightarrow Y'$ identifies Y' with $\tilde{Z}_4$. Moreover, an elementary transformation at r_4 gives an explicit seven-point blow-up presentation $\sigma_4 : \tilde{Z}_4 \rightarrow \mathbb{F}_2$. Its centers, in order, are the transported points $p_2, p_3, p_4, p_5, r_1, r_2, r_3$: the first five are distinct on $\mathbb{F}_2$, and the last two are infinitely near p_2 .

Proof. All divisor classes on Y in the calculations below are total pullbacks, except where strict transforms are explicitly specified. The four (-1) -curves C_i form a cycle with distinct vertices, so their strict transforms

$$E_i = C_i - R_{i-1} - R_i$$

satisfy $E_i^2 = -3$ and $E_i E_j = 0$ for $i \neq j$. The contraction $Y \rightarrow X_{4,4/3}$ is the 4-gon construction of [11, Section 3.5].

The conic through five general points is smooth. Its tangent line at p_1 contains none of $p_2, \dots, p_5$, by Bézout, and meets Q_1 at r_4 after the first blow-up. This point is different from r_1 , the direction of the secant through p_2 . Thus D is the strict transform of an irreducible smooth rational curve. In the orthogonal blow-up basis we have

$$D^2 = -1, \quad (-K_Y)D = 1, \quad DE_i = 0 \quad (1 \leq i \leq 4).$$

These distinct irreducible curves are therefore disjoint. The image of D on $X_{4,4/3}$ lies in the smooth locus and is a floating (-1) -curve. Its contraction is a log del Pezzo surface with four unchanged $\frac{1}{3}(1,1)$ -points and degree $7/3$, hence is Z_4 by the classification in Section 5. Contracting D on Y leaves the four E_i unchanged and gives its minimal resolution Y' .

To identify the starting ruled surface, first blow up only p_1 . This gives $\mathbb{F}_1$, with negative section Q_1 and ruling class $H - Q_1$. The tangent direction of the conic specifies r_4 on Q_1 . Put $W = \text{Bl}_{r_4} \mathbb{F}_1$. The strict transform of that ruling fiber is D . Contracting it is the elementary transformation of the ruled surface at a point on its negative section. On $\mathbb{F}_1$, we have $Q_1^2 = -1$. Since r_4 lies on Q_1 , its strict transform on W has square -2 . It is disjoint from D , so contracting D does not change this self-intersection. The descended ruling therefore has a section of square -2 , and the classification of ruled surfaces identifies the result with $\mathbb{F}_2$ [18, Chapter V, Section 2].

It remains to account for the marked centers. On W , perform the blow-ups at p_2, p_3, p_4, p_5 , then at r_1, r_2, r_3 . All these centers, including the two directions r_2, r_3 on the exceptional curve over p_2 , lie away from the strict transform of D : the tangent line avoids the four points, and $r_1 \neq r_4$. The isomorphism

$$W \setminus D \simeq \mathbb{F}_2 \setminus \{t\},$$

where t is the image of D , transports these centers and the points infinitely near to p_2 to the corresponding blow-up of $\mathbb{F}_2$. Blow up the images of p_2, p_3, p_4, p_5 , followed by the transported r_1, r_2, r_3 . The transported r_1 lies on the negative section of $\mathbb{F}_2$. Thus the transported sequence gives the asserted morphism $\sigma_4 : \tilde{Z}_4 \rightarrow \mathbb{F}_2$. $\square$

Using precisely this order of seven centers in (9), we obtain

$$\mathcal{A}_4 = \langle G_7, \dots, G_1, \sigma_4^* \mathcal{O}, \sigma_4^* \mathcal{O}(f), \sigma_4^* \mathcal{O}(C_0 + 2f), \sigma_4^* \mathcal{O}(C_0 + 3f) \rangle.$$

Here C_0 is the negative section on the new $\mathbb{F}_2$.

C.5 The five-point base

For the five-point base, the four plane blow-ups followed by the five vertex blow-ups of the pentagon on the degree-five del Pezzo surface give a marked blow-up presentation of $\tilde{Z}_5$ over the plane and hence its Orlov collection. On the degree-five del Pezzo surface $S_5 = \text{Bl}_{p_1, \dots, p_4} \mathbb{P}^2$, take the pentagon from Appendix B.3:

$$\begin{aligned} C_1 &= Q_1, & C_2 &= H - Q_1 - Q_2, & C_3 &= Q_2, \\ C_4 &= H - Q_2 - Q_3, & C_5 &= H - Q_1 - Q_4. \end{aligned}$$

Each curve has square -1 , adjacent curves intersect once, and non-adjacent curves are disjoint. Blow up the five distinct vertices $r_i = C_i \cap C_{i+1}$, with exceptional curves R_i . Since exactly two vertices lie on each C_i , the strict transforms

$$E_i = C_i - R_{i-1} - R_i$$

have square $-1 - 2 = -3$. Blowing up the unique intersection of any adjacent pair separates it, so $E_i E_j = 0$ for $i \neq j$. By the 5-gon construction of [11, Section 3.5], their contraction belongs to $X_{5,5/3}$. Uniqueness of this base identifies the contraction with Z_5 , and consequently

$$\tilde{Z}_5 \simeq \text{Bl}_{r_1, \dots, r_5} \text{Bl}_{p_1, \dots, p_4} \mathbb{P}^2.$$

Here the vertex blow-ups are on S_5 , not at nine distinct points of the plane: r_1, r_5 lie on Q_1 over p_1 , r_2, r_3 lie on Q_2 over p_2 , and r_4 maps to a fifth distinct plane point. After each vertex blow-up, the remaining vertices are taken on the strict transforms of the indicated curves. Order the centers as $p_1, \dots, p_4, r_1, \dots, r_5$, and write σ_5 for their composite. The twelve-object Orlov collection is

$$\mathcal{A}_5 = \langle G_9, \dots, G_1, \sigma_5^* \mathcal{O}, \sigma_5^* \mathcal{O}(H), \sigma_5^* \mathcal{O}(2H) \rangle.$$

C.6 The six-point base

For the six-point base, we contract six disjoint toric boundary curves of $\tilde{Z}_6$ to obtain the degree-six del Pezzo surface, then two opposite boundary curves to obtain $\mathbb{F}_0$. Reversing these contractions yields the twelve-object exceptional collection on the resolution, despite the Brauer obstruction on the singular surface. Use the fan and notation of Appendix B.4. The resolved boundary is $V_1, E_1, \dots, V_6, E_6$, with $V_i^2 = -1$ and $E_i^2 = -3$. We first contract $V_1, \dots, V_6$, in that order. At every stage, the neighbors of the ray v_i being removed are still u_{i-1}, u_i , and

$$u_{i-1} + u_i = v_i, \quad \det(u_{i-1}, u_i) = 1.$$

Thus every contraction is a smooth toric blow-down. Equivalently, the V_i are pairwise disjoint, so their self-intersections remain -1 until they are contracted. After any initial set I of these contractions, the self-intersection of the image of E_j is

$$-3 + \#(I \cap \{j, j+1\}),$$

with cyclic indices. After all six, the fan consists of the u_i and every boundary curve has square -1 . This is the smooth toric del Pezzo surface of degree six: its anticanonical divisor intersects each invariant curve positively, and its square is six.

Next contract the opposite boundary curves with rays u_1 and u_4 . They are disjoint, and their ray relations are

$$u_1 = u_6 + u_2, \quad u_4 = u_3 + u_5, \quad \det(u_6, u_2) = \det(u_3, u_5) = 1.$$

Both are therefore (-1) -curves at the moment of contraction. The remaining rays are $u_2, u_3, -u_2, -u_3$, and $\det(u_2, u_3) = 1$. In this lattice basis the fan is the square fan of $\mathbb{F}_0 = \mathbb{P}^1 \times \mathbb{P}^1$.

Reversing the sequence prescribes eight toric point blow-ups: first insert u_4 between u_3, u_5 , then u_1 between u_6, u_2 , and then insert $v_6, v_5, \dots, v_1$, each between u_{i-1}, u_i . The insertions of u_4, u_1, v_6, v_3 blow up the four original fixed points of $\mathbb{F}_0$; the remaining four

blow-ups are at points infinitely near to those producing u_4 and u_1 . The insertions of v_5, v_4 have centers on the successive strict transforms of the curve with ray u_4 , and those of v_2, v_1 on the successive strict transforms of the curve with ray u_1 . Let $\sigma_6 : \tilde{Z}_6 \rightarrow \mathbb{F}_0$ be their composite. With f_1, f_2 the two ruling classes on $\mathbb{F}_0$, the resulting twelve-object collection is

$$\mathcal{A}_6 = \langle G_8, \dots, G_1, \sigma_6^* \mathcal{O}, \sigma_6^* \mathcal{O}(f_1), \sigma_6^* \mathcal{O}(f_2), \sigma_6^* \mathcal{O}(f_1 + f_2) \rangle.$$

The marked blow-up presentations in [Appendices C.1 to C.6](#) establish [Proposition 5.4](#). The following corollary extends the resulting exceptional collections along the blow-ups at distinct smooth points of Z_k described in [Section 5](#), proving [Theorem 1.1](#). We use $A_j^{(k)}$ for the entries of $\mathcal{A}_k$ on $\tilde{Z}_k$, reserving G_i for blow-up objects.

Corollary C.3. *Let X belong to a Fano index one family $X_{k,d}$, and put $s = K_{Z_k}^2 - K_X^2$. Choose a marked cascade presentation by s distinct smooth points of Z_k , and let $\gamma : \tilde{X} \rightarrow \tilde{Z}_k$ be the induced map of minimal resolutions. Write the exceptional collection on $\tilde{Z}_k$ as*

$$\mathcal{A}_k = \langle A_1^{(k)}, \dots, A_{N_k}^{(k)} \rangle,$$

with N_k as in [Table 1](#), and let $D_1, \dots, D_s$ be the disjoint exceptional curves of γ . Set $H_i = \mathcal{O}_{D_i}(-1)$; the distinctness of the centers follows from [\(3\)](#). The derived-pullback convention [\(9\)](#) is needed only for the collections on $\tilde{Z}_k$, whose marked blow-up presentations may use points infinitely near to earlier points as blow-up centers. Then

$$D^b(\mathrm{coh} \tilde{X}) = \langle H_s, \dots, H_1, L\gamma^* A_1^{(k)}, \dots, L\gamma^* A_{N_k}^{(k)} \rangle$$

is a full exceptional collection. If $\mathcal{X}$ is the canonical stack and Φ its Ishii–Ueda embedding, the ordered collection

$$D^b(\mathrm{coh} \mathcal{X}) = \langle \mathcal{E}_{p_1}, \dots, \mathcal{E}_{p_k}, \Phi(H_s), \dots, \Phi(H_1), \Phi(L\gamma^* \mathcal{A}_k) \rangle$$

is full exceptional, of length $N_k + s + k = 12 - K_X^2 + 4k/3$, the specialization of [Corollary 3.6](#) with $n_i = 3$ and $r = k$.

Remark C.4 (Marked presentations and dominance). Every individual member admits the blow-up presentation of [\[11, Corollary 8\(1\)–\(6\)\]](#), with distinct smooth centers by [\(3\)](#). Applying Orlov’s formula and the Ishii–Ueda embedding gives all the collections in [Theorem 1.1](#). [Appendix A](#) compares the parameter space of ordered eight-point configurations on $\mathbb{P}(1, 1, 3)$ with the parameter space of degree-10 hypersurfaces in $\mathbb{P}(1, 2, 3, 5)$. Its automorphism and fiber calculations show that general hypersurfaces admit general centers, and general centers give general hypersurfaces.

For other families, an analogous comparison with a specified equation parameter space or an alternative marked blow-up presentation requires its own classifying map and fiber analysis. The infinitely near points used as blow-up centers in [Appendix C](#) occur only in the fixed marked blow-up presentations of $\tilde{Z}_k$, not in the additional blow-ups at distinct smooth points of Z_k .

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used ChatGPT (OpenAI) to assist with language editing, organization, and manuscript formatting. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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